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THE EFFECTS OF COMPETITION IN THE RETAIL GASOLINE INDUSTRY

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The Effects of Competition in the Retail Gasoline Industry
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ABSTRACT

We estimate the effect of competition on incumbent firm pricing by using high-frequency price data and the precise geographic location for all gas stations in California. Using an event-study design, we find that the entry of a new station is associated with an immediate, persistent 2.7-cent decrease in prices at incumbent stores, which equates to a 7% reduction in estimated retail markups. In contrast, nearby exit results in precisely estimated null effects on prices. We do not find that observable differences in station characteristics explain the asymmetry. Rather, we find that selective exit from more competitive markets and less heavily-trafficked stations explains the null results for exit. We discuss the relevance of our estimates to current environmental policy related to the decarbonization of light-duty transportation and zoning and environmental restrictions on gasoline stations.

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1 Introduction

On-road gasoline and diesel fuel accounts for roughly one-quarter of U.S. carbon emissions. While much of the policy debate and academic research around transportation decarbonization explores the role of policies to shift fuel use (e.g., fuel taxes and clean fuel standards) or the composition of the vehicle fleet (e.g., electric vehicle subsidies and mandates), the structure of the retail fuel market plays an important role in the pricing and availability of different fuels. In turn, this impacts the attraction of different light-duty vehicle power-trains and the carbon-intensity of the sector.

Although vehicle miles traveled reached all-time highs in 2024, gasoline demand has not rebounded to pre-pandemic levels and is projected to decline over the next several decades as fuel efficiency improves and electric vehicle penetration increases.¹ Against the backdrop of falling expected demand, local governments increasingly exercise zoning and regulatory authority to constrain station entry and speed station exit. Starting with Petaluma in 2021, a number of cities in California have banned the construction of new gas stations and placed restrictions on the expansion of existing stations. Understanding how entry and exit impact retail gasoline prices is central to understanding both the welfare and distributional consequences of decarbonization policy.

In this paper, we estimate the effects of station entry and exit in local markets for retail gasoline in California, a state that faces both high gasoline prices and is a leader in both transportation decarbonization and local zoning restrictions. To calculate the reduced-form causal effect, we use a panel of daily station-level prices and the precise geographic location for the universe of gas stations from 2014-2018, and exploit geographic and temporal variation in exposure to nearby entry and exit. The resulting data set includes over 700 new station entry and exit events with 35 million price observations. In the spirit of [Arcidiacono et al. \(2020\)](#), we use both difference-in-differences and event-study designs to compare the change in prices at incumbent stations before and after nearby entry or exit to price changes at stations not exposed to such events. Given the

¹see, e.g., Energy Information Administration, Annual Energy Outlook 2025, Table 37.

high-frequency nature of the data, we are able to control for the endogenous location decision of entering and exiting firms with a rich panel of station and day-of-sample fixed effects and city-specific linear time trends. Under the assumption that the exact timing of the entry or exit events is conditionally exogenous, the coefficient is identified by within-station variation in the number of nearby competitors. Our event-study results support this key identifying assumption – stations with and without nearby entry or exit follow parallel trends in the periods leading up to the respective event. In addition, we find little evidence of turnover, when the effect of entry or exit is attenuated by entry or exit of other stations in the periods preceding, following, or simultaneous with the original entry and exit.

We find that increasing nearby competitors reduces prices at incumbent stations, but that the effects from entry and exit are asymmetric. Entry of a new gas station is associated with a 2.7 cent reduction in gas prices at nearby (less than 1 mile away) incumbent stations, representing a 7% reduction in average retail markups. Our event-study specification highlights that the effect of entry is immediate, occurring at the time of the entry event, and persistent, lasting for years. It is present for all grades of gasoline and attenuates with the distance between the incumbent stations and the entrant, consistent with prior literature on the tight spatial nature of retail gasoline competition. Lastly, we find suggestive evidence of heterogeneity across types of stations. Notably, high-volume hypermart stations (i.e. Costco) have a stronger entry effect on incumbents than branded and unbranded entrants.

In sharp contrast, we estimate precise null effects of station exit on nearby incumbent station pricing. We explore this asymmetry by comparing the nature of entry and exit events. We find little evidence that the asymmetry is attributable to observable station characteristics, such as the location of the station or station branding, or how entry and exit events are identified in the data. But we do find evidence that entry tends to occur in areas with more concentrated stations, where we estimate the largest impacts from entry. In addition, we find suggestive evidence that exiting stations differ from incumbent (or entering) stations on unobservable dimensions, as reflected by the frequency with which crowd-sourced prices are reported pre-exit. Taken together, we find that

the exit of a highly-competitive station in markets with few competitors leads to a significant price increase, similar in magnitude to that of a station entry. We conclude that the observed asymmetry is driven by differences in the locations where exit occurs and the unobservable characteristics of the exiting station.

Our work contributes to a recent, growing literature that uses station-level data to estimate the effect of market structure on retail fuel prices. Although there has long been interest in entry and exit in retail fuel markets, (e.g., [Barron et al. \(2004\)](#); [Tappata and Yan \(2017\)](#)), estimation in the spirit of [Arcidiacono et al. \(2020\)](#) has, until recently, been limited by a lack of granular price data in settings with sufficient changes in market structure ([Haucap et al. 2017](#)). Here, our work complements a growing series of papers that estimate the effect of station entry in Spain ([Bernardo 2018](#); [González and Moral 2023](#)), Germany ([Fischer et al. 2025](#)), Mexico ([Davis et al. 2022](#)) and Australia ([Ormosi et al. 2025](#)) using high-frequency administrative data. We contribute to this literature in two respects.

First, recent papers focus almost exclusively on the impact of station entry on prices. Our paper provides some of the first evidence of the effect of both entry and exit events and is the first (to our knowledge) to highlight the asymmetry in the impacts of station entry and exit on incumbent prices.² Our focus on the asymmetric response to entry and exit complements the recent work by [Ormosi et al. \(2025\)](#), who study entry and exit effects in Western Australia, but focus on how the magnitude of entry and exit events relates to the income level of the local market.

The asymmetry of the impact of entry and exit is particularly relevant in light of the broader energy transition away from fossil fuels in the transportation sector. Quantifying how markets will be impacted by a shift towards electrified transportation is central to understanding the dynamics of the energy transition and the distributional consequences for groups that continue to use vehicles powered by gasoline. Here, our paper offers some preliminary insight into the potential short-term effects of market structure on gasoline prices should the energy transition lead to falling demand

²Although we are not aware of other papers that focus on the asymmetric effects of entry and exit in retail gasoline, the asymmetric speed with which gasoline prices rise and fall, termed rockets and feathers, has long been studied. (see, e.g., [Borenstein et al. \(1997\)](#); [Lewis and Noel \(2011\)](#))

for liquid transportation fuels and modest station exit. If, as our findings suggest, initial exit from the market is largely driven by less-competitive stations in higher density markets, the initial wave of exits might have relatively modest effects on incumbent station pricing. But, if changing market conditions or entry restrictions (such as those in Petaluma) induce exit or limit entry of larger, more competitive stations, these impacts would tend to increase prices. Moreover, exit may impose additional welfare costs by reducing local station density and increase travel or waiting costs for drivers.

Second, our work provides the first event-study-based causal estimates of the effect of station entry and exit on pricing in California. As such, our work contributes to the understanding of competition in the California gasoline market. For the past two decades, gasoline in California has been significantly more expensive than gasoline in the rest of the country, after accounting for higher excise taxes, state-specific environmental regulations³, and high entry costs due to zoning laws and high land values. This gap has drawn scrutiny from academics (e.g., [Borenstein et al. \(2004\)](#) examine wholesale market power; [Hastings \(2004\)](#); [Taylor et al. \(2010\)](#) examine vertical mergers, among others) and spurred investigations from the California Energy Commission⁴ and the California Department of Justice⁵. In particular, the California Energy Commission report concludes that over the last several decades, “the primary cause of the residual price increase is simply that California’s retail gasoline outlets are charging higher prices”. Although we find that incumbents lower their prices after the entry of a new competitor, the effect sizes are modest relative to the residual price gap of 30-40 cents during this time period as estimated by the CEC.

³For three decades, stations in California, must sell a special blend of gasoline (CARBOB), distinct from the fuel sold in surrounding states. California’s unique gasoline blend regulations are served almost exclusively by the few in-state refineries.

⁴https://www.energy.ca.gov/sites/default/files/2025-07/Gas_Price_Report_ada.pdf

⁵<https://oag.ca.gov/antitrust/gasoline>

2 Data

To estimate the effect of station entry and exit on incumbent pricing, we use daily station-level gasoline prices from the Oil Price Information Service (OPIS) for all California stations from 2014–2018. We observe precise latitude and longitude coordinates and branding for each station. Throughout the paper, we use the term “station” to refer to the unique site identifier in OPIS. This identifier remains fixed as a station undergoes renovations or branding changes, allowing us to distinguish true entry and exit from within-station changes that do not change the number of competitors.

Following the California Energy Commission, we classify stations into three branding categories.⁶ “Branded” stations sell gasoline under the brand of a refining company and typically include proprietary fuel additives.⁷ “Unbranded” stations sell gasoline under non-refiner brands and typically purchase fuel at the wholesale rack (terminal) without exclusive supply contracts. Finally, “hypermarts” are high-volume retailers such as Costco, Walmart, and Safeway, which also typically sell unbranded fuel.

The data include 9,716 stations in California with more than 35 million price observations across the three major gasoline grades: regular, mid-grade, and premium.⁸ OPIS collects prices from monitored fleet credit card transactions, customer reports, and direct station feeds, and reports one price per station per day for each grade. Coverage is high: the median station reports 1,756 daily prices (96% of days), and 75% report more than 1,500 observations (82% of days) over the sample period. We exclude stations with fewer than 14 price observations over the five-year period to avoid biasing our entry and exit designations and one station lacking geographic data, yielding a final sample of 9,539 stations.⁹

⁶https://www.energy.ca.gov/sites/default/files/2020-02/2020-01_Petroleum_Watch.pdf

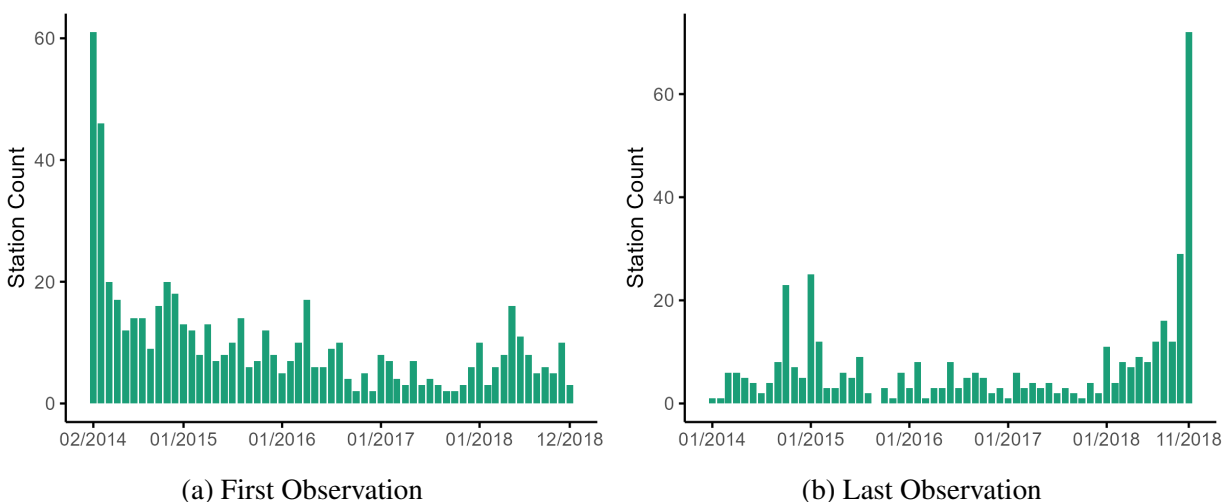
⁷Specifically Chevron, Shell, 76, Exxon Mobil, and Valero are classified as “branded” during our sample period. Although we observe the retail brand, we do not observe whether the station is independently owned, franchised, directly owned by a petroleum company, or operates under a consignment arrangement.

⁸Regular, mid-grade, and premium correspond in California to unleaded gasoline with octane levels of 87, 89, and 91, respectively. Regular gasoline represents roughly 70% of motor gasoline sales in California, premium about 25%, and mid-grade the remainder.

⁹Appendix B provides additional details on data construction and entry/exit designation.

To our knowledge, there is no comprehensive public source of gas station operating dates in California.¹⁰ We therefore leverage the high frequency of reporting in the OPIS data to identify entry and exit. Conceptually, we classify stations based on the first and last observed price dates.¹¹

Figure 1: Month of First and Last Observation, by Station



Notes: Month of first and last observation for each station. January 2014 is removed for panel (a) and December 2018 is removed from panel (b) for scale. Data are sourced from the Oil Price Information Service (OPIS) for all stations in California for 2014-2018.

Figure 1 graphs the number of stations by month of the first and last observation. Panel (a) excludes January 2014 and panel (b) excludes December 2018 since the overwhelming majority of first and last observations fall in these two months, as to be expected. To avoid mechanical misclassification of stations as entrants or exits at the sample boundaries due to intermittent reporting, we do not consider entry in January–March 2014 and exit in November–December 2018 as events.¹²

The main concern with previous cross-sectional analyses is that entry and exit are selected and likely to occur in systematically different markets. Failure to account for these differences can lead to omitted variable bias. To assess baseline differences, we combine demographic data from the 2014 American Community Survey (ACS) 5-year estimates, station characteristics from OPIS

¹⁰Business permits and underground storage tank data can identify station locations but generally lack reliable operating dates. State franchise tax records are reported at the owner level and do not provide store-level timing. California Energy Commission surveys are voluntary and annual, lacking precise entry and exit dates.

¹¹In this design, stations that go dormant for a period but later resume operations are not considered an exit and subsequent entry.

¹²Our results are robust to more restrictive definitions as shown in Appendix B.2.

and the California Energy Commission's A15 census, and state vehicle registration data.

Table 1: Baseline Differences, by Entry and Exit Status

	Entry	Exit	Difference
<i>Station Characteristics</i>			
Station Count	484	348	
Branded Gas	0.250 (0.433)	0.345 (0.476)	-0.095 (0.003)
Hypermart	0.052 (0.222)	0.023 (0.150)	0.029 (0.026)
# of Competitors	3.479 (3.094)	4.569 (3.259)	-1.090 (0.000)
Distance to Highway	1,323 (3,041)	796 (1,423)	527 (0.001)
< .25 Mile to Highway	0.510 (0.500)	0.563 (0.497)	-0.053 (0.131)
Service Bay	0.013 (0.114)	0.065 (0.247)	-0.052 (0.006)
Car Wash	0.160 (0.367)	0.119 (0.325)	0.041 (0.191)
Convenience Store	0.660 (0.474)	0.662 (0.474)	-0.002 (0.971)
Kiosk	0.049 (0.216)	0.060 (0.238)	-0.011 (0.608)
Restaurant	0.082 (0.274)	0.060 (0.238)	0.022 (0.338)
Supermarket	0.036 (0.186)	0.035 (0.184)	0.001 (0.947)
<i>Market Characteristics</i>			
BEV Penetration	0.002 (0.003)	0.004 (0.005)	-0.001 (0.000)
Gas Price	3.958 (0.152)	3.975 (0.178)	-0.016 (0.210)
Income (Median)	55,182 (24,118)	55,626 (25,880)	-444 (0.811)
Poverty Rate	19.912 (12.403)	19.996 (13.100)	-0.084 (0.929)
Households	1,957 (958)	1,921 (922)	36 (0.600)
House Value (Median)	275,643 (177,797)	330,559 (228,081)	-54,916 (0.000)
Population Density	1,497 (1,855)	1,886 (2,227)	-388 (0.011)
% No Vehicle	7.024 (6.513)	8.381 (8.210)	-1.358 (0.015)
% Commuting by Vehicle	86.245 (10.509)	84.977 (11.208)	1.268 (0.116)
Commute Time	26.812 (7.112)	25.495 (6.113)	1.317 (0.007)

Notes: Means and standard deviations are reported in Columns 1 and 2. The difference in means is shown in Column 3 with the associated p-value reported below. BEV penetration is reported by zip code while the remainder of market characteristics are reported at the census tract level. Data are sourced from the 2014 American Community Survey 5-year estimates, the Oil Price Information Service, and the California Energy Commission.

In Table 1, we compare entry and exit by reporting summary statistics for station characteristics (top panel) and local area characteristics (bottom panel).¹³ Focusing first on the station characteristics, entering stations are more likely to be unbranded, hypermarts, and face fewer competitors than exiting stations. In terms of station amenities, entrants have fewer attached service bays for repairs, but more car washes and restaurants.

Gasoline prices are indistinguishable over the first 3 months of our sample. While the sample is early in the adoption curve of electric vehicles, exit markets have higher BEV penetration. Entry and exit tracts have similar household income, poverty rates, and housing density. However, exit tracts have higher population density and higher house values. Focusing on related driving characteristics, tracts experiencing entry have lower shares of households without vehicles, longer average commute times, and more commuting by car.

3 Empirical Strategy

The localized nature of gasoline station competition allows geographic and temporal variation in exposure to nearby entry and exit to identify causal effects on incumbent pricing in a difference-in-differences framework. Following Arcidiacono et al. (2020), we treat the exact timing of station entry and exit as plausibly exogenous short-run shifts to local market structure for incumbent firms after conditioning on the inherent market structure.

Importantly, we include a rich panel of fixed effects to mitigate bias from unobserved variables inherent to the endogenous entry and exit decisions. By restricting the model to identification from within-station variation in the number of nearby competitors, over time, the specification absorbs time-invariant factors that may influence both location decisions and pricing such as the overall price level in the market, local price elasticity of demand, local traffic patterns, and relevant customer characteristics.

Formally, we estimate:

¹³Battery Electric Vehicle (BEV) penetration is reported at the zip code level from the data source while the remainder of the local market characteristics are reported at the tract level.

$$P_{st} = \alpha + \beta N_{st} + \sigma_s + \delta_t + \Phi_c(t) + \varepsilon_{st} \quad (1)$$

where the main outcome variable is the retail price in nominal dollars per gallon at station s on day t and N_{st} is the count of competitors nearby station s on day t , increasing upon entry and decreasing with nearby exit. In our preferred specifications, we define the relevant potential competitors as those within a 1-mile radius circle. This is in line with the prior literature that uses short distances, driving distances, and driving times to determine proximity (Lewis 2015; Davis et al. 2022; Fischer et al. 2025; Hastings 2004; Bernardo 2018; Carranza et al. 2015; Barron et al. 2004).

Station fixed effects (σ_s) capture time-invariant differences across locations, such as station amenities, size, and proximity to wholesale terminals. Day-of-sample fixed effects (δ_t) capture state-wide daily shocks to both input costs (e.g., oil prices and refinery outages) and common demand conditions. In preferred specifications, we also include city-specific linear time trends, $\Phi_c(t)$, to flexibly account for differential local trends that may be correlated with prices and station demand. We cluster standard errors at the city level to account for common shocks across units.

The specification above constrains the impact of entry and exit on prices to be the same magnitude (although of opposite sign). To relax this assumption, we also estimate the following difference-in-differences specification:

$$P_{st} = \alpha + \beta_1 \text{Entry}_{st} + \sigma_s + \delta_t + \Phi_c(t) + \varepsilon_{st}, \quad (2)$$

where Entry_{st} is an indicator equal to one beginning in the month of the first nearby entry event experienced by station s and zero beforehand. We estimate the exit effect analogously using an indicator for the first nearby exit event.

Equations (1) and (2) are two-way fixed effects (TWFE) estimators in which never-treated, not-yet-treated stations, and previously-treated stations serve as controls. To address potential bias in staggered treatment settings with heterogeneous effects, we implement the robust estimator of

Gardner et al. (2024).

Identification of the main coefficient of interest, β_1 , requires that, conditional on the included fixed effects and trends, the timing of nearby entry and exit is as-good-as random with respect to unobserved determinants of incumbent pricing:

$$E[\varepsilon_{st} \mid \sigma_s, \delta_t, \Phi_c(t), N_{st}] = 0. \quad (3)$$

This requires that there were no other factors correlated with the timing of the station entry or exit that also impacted the pricing of nearby stations. Although this assumption cannot be directly tested, demonstrating that the treatment and control follow common trends in the pre-treatment period offers a falsification test of the assumption. Secondly, the difference-in-differences framework assumes stable unit treatment values which requires that there is no spillover of treatment onto control units outside of the impacted market. This assumption is plausible in our setting given the highly localized nature of gasoline competition, which limits spillovers beyond the defined geographic market.

To examine the price dynamics of entry and exit and to test pre-trend validity, we estimate the following event-study specification for station entries, and the equivalent analog for exits:

$$P_{st} = \alpha + \sum_{k=-24}^{24} \beta_k [Entry_{st} = k] + \sigma_s + \delta_t + \Phi_c(t) + \varepsilon_{st} \quad (4)$$

where event time indicators capture the number of months before and after the first nearby entry (or exit) experienced by station s . Endpoints are binned to include 24 or more months before and after the event.

The event-study framework complements the static difference-in-differences estimates. Notably, precisely estimated null effects in the time periods leading up to entry or exit provide support that treatment and control markets were following common trends. In addition, the event-study specification allows for the effect of entry or exit to evolve dynamically, illustrating whether prices change quickly or gradually following a change in the competitive landscape. As before, we es-

timate the event studies using the estimator from [Gardner et al. \(2024\)](#) to recover the average treatment effect.

4 Results

We present our estimates for the effect of entry and exit on incumbent pricing in Table 2. The top panel reports results from estimation of equation (1), which relates incumbent prices to the number of competitors within 1 mile. The bottom two panels report estimates from equation (2), which separately identifies the effects of entry and exit. Moving from left to right, Column 1 reports estimates from a simple linear specification of nominal gasoline prices on the variable of interest. Columns 2–4 progressively add station fixed effects and day-of-sample fixed effects. Column 5 further incorporates city-specific linear time trends. Column 6 presents estimates using the two-stage difference-in-differences estimator ([Gardner et al. 2024](#)).

Table 2: Effect of Station Entry and Exit Within 1 Mi. on Incumbent Pricing

	(1)	(2)	(3)	(4)	(5)	(6)
Station Count	-0.006*** (0.002)	-0.067*** (0.021)	-0.005** (0.002)	-0.012*** (0.003)	-0.010*** (0.003)	
R Sq.	0.001	0.163	0.809	0.956	0.958	
Obs.	14,749,777	14,749,777	14,749,777	14,749,777	14,749,777	
Entry	-0.128*** (0.011)	-0.310*** (0.023)	-0.044*** (0.010)	-0.023*** (0.004)	-0.020*** (0.004)	-0.025*** (0.004)
R Sq.	0.005	0.170	0.809	0.956	0.958	
Obs.	14,749,777	14,749,777	14,749,777	14,749,777	14,749,777	14,748,218
Exit	-0.066*** (0.014)	-0.193*** (0.030)	-0.003 (0.011)	0.002 (0.004)	0.001 (0.004)	0.000 (0.005)
R Sq.	0.001	0.165	0.808	0.956	0.958	
Obs.	14,749,777	14,749,777	14,749,777	14,749,777	14,749,777	14,749,777
Station FE	No	Yes	No	Yes	Yes	Yes
Day of Sample FE	No	No	Yes	Yes	Yes	Yes
Linear Time Trend	No	No	No	No	Yes	No
Estimator	TWFE	TWFE	TWFE	TWFE	TWFE	DID2S

Notes: The table reports estimates for the effect of a change in nearby competitors within 1-mile on incumbent pricing for regular unleaded gasoline in nominal dollars per gallon. The top panel uses the number of nearby stations as the independent variable. The panels for Entry and Exit use an indicator variable for the first nearby entry or exit. Column 1 reports estimates from a simple linear regression. Columns 2-5 report results with the addition of the fixed effects listed in the panel below. Column 6 uses the two-stage DID estimator from [Gardner et al. \(2024\)](#). Data are sourced from the Oil Price Information Service (OPIS) for all stations in California for 2014-2018. Model standard errors are reported in parentheses and clustered at the city level. *** = significant at 1 percent level, ** = significant at 5 percent level, * = significant at 10 percent level.

In the top panel, we estimate a consistently negative relationship between incumbent prices and the number of nearby competitors. Across specifications, with the exception of Column 2, the estimates fall in a narrow range: an additional competitor is associated with incumbent prices that are between 0.5 to 1.2 cents per gallon lower.¹⁴ Column 4 presents the canonical two-way fixed effects estimate. In this specification, an additional competitor reduces prices by roughly 1.2 cents per gallon. The estimate remains stable with the inclusion of city-specific linear time trends in Column 5. Economically, this effect represents about a 2.5% reduction in the average station gross markup of 40 cents per gallon, as reported by the California Energy Commission.¹⁵

¹⁴The difference in Column 2 arises since station density tended to increase and prices tended to fall over the study period, biasing the coefficients in Column 2 downwards relative to the two-way fixed effects estimates in Column 4.

¹⁵The California Energy Commission provides estimates of gross retail margins calculated as the difference between average retail prices and wholesale gasoline prices, less applicable state and local taxes and fees. This does not

These estimates control for potential selection if entry or exit are correlated with local demand or price trends. The robustness of the coefficient to the inclusion of a city-specific linear time trend provides support that treatment and control markets are not trending differentially.

In the second and third panels of Table 2, we estimate regressions of incumbent prices on indicators for the period following the first nearby entry or exit. We focus on Columns 5 and 6, which control for both time-invariant and station-specific unobservables and identify the coefficient from within-station variation relative to statewide trends in prices. In our preferred specification, incumbent prices fall by 2.5 cents per gallon following the entry of a new competitor within 1 mile. In contrast, we detect no statistically meaningful price response following competitor exit, yielding a precisely estimated null effect. By breaking out the entry and exit effects separately, we can see that the prior results using station count as the regressor attenuated the effect due to the asymmetry by event type. In Column 6 we show that the entry and exit estimates are also robust to implementation of the estimator in [Gardner et al. \(2024\)](#). We use this estimator in the analyses that follow.

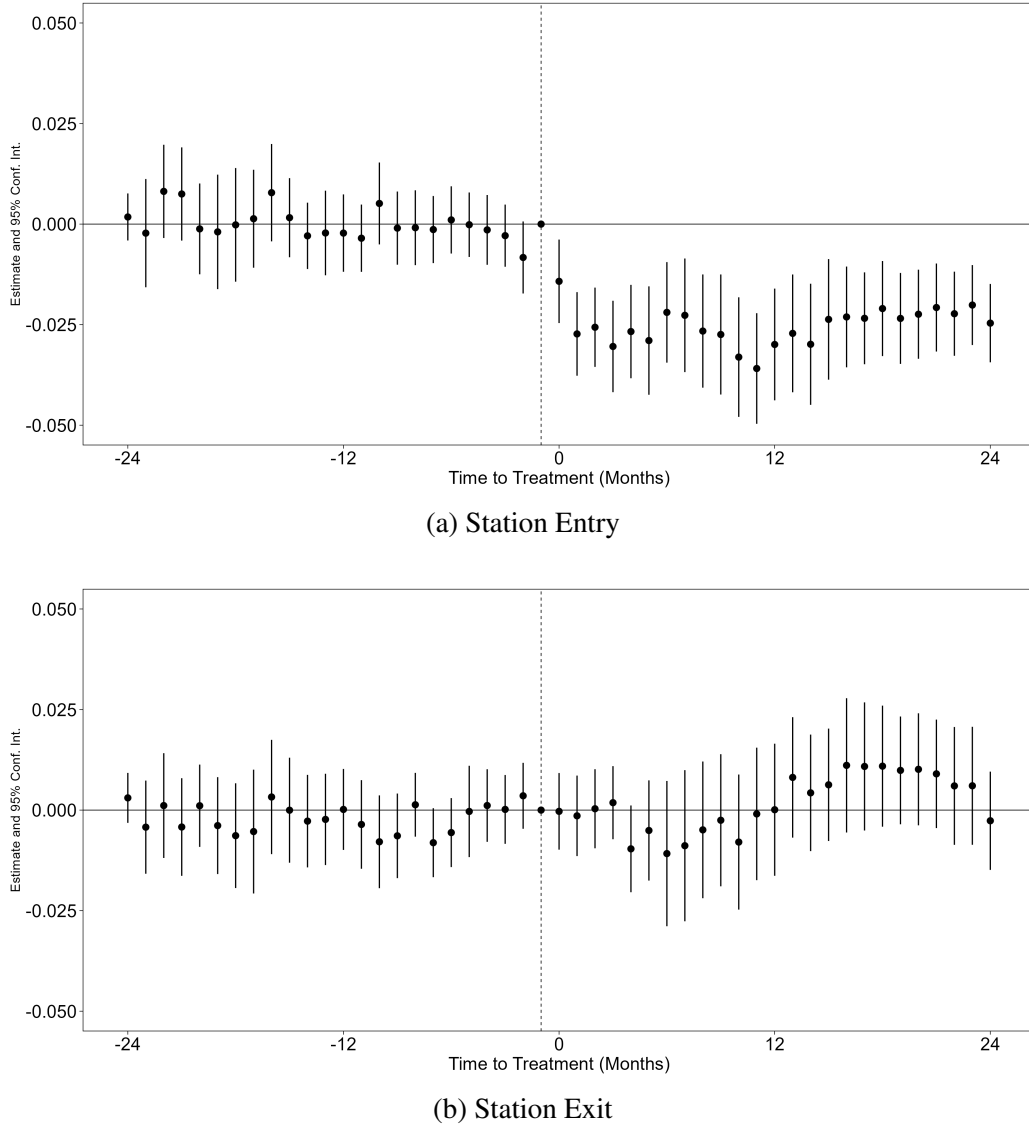
The event-study specification, formalized in equation (4), allows us to examine the timing and persistence of incumbent price responses following nearby entry or exit. In addition, the event-study design allows us to visually and statistically assess the parallel trends assumption for the pre-treatment period. Evidence of no differential pre-trends supports the identifying assumption that, absent treatment, prices would have evolved similarly across the two groups.

Figure 2a presents event-study estimates for the first entry event experienced by each incumbent station, while Figure 2b presents the corresponding estimates for the first exit event during the sample period.¹⁶

account for any station specific marketing or operating expenses.

¹⁶We compare event-study coefficients from the TWFE and two-stage DID estimators in Appendix Figure A1. Results are similar between the two estimators.

Figure 2: Effect of Station Entry and Exit Within 1 Mi. on Incumbent Pricing



Notes: Graph plots the coefficient estimates and 95% confidence intervals for the effect of station entry and exit within 1 mile on incumbent pricing of regular unleaded gasoline. Regressions are estimated using the two-stage DID estimator in [Gardner et al. \(2024\)](#). Standard errors are clustered at the city level. Data are sourced from the Oil Price Information Service (OPIS) for all stations in California for 2014-2018.

Examining the pre-treatment point event-time estimates in the months leading up to treatment, all periods are close to zero and include zero within the confidence intervals. The absence of pre-trends before entry provides evidence that incumbents do not systematically lower prices in anticipation of nearby entry. Similarly, we find no evidence of price movements prior to exit that would be consistent with incumbents strategically lowering prices to induce competitor exit.

Turning to the post-treatment coefficients for entry, Figure 2a shows that incumbent prices fall discretely and immediately following the arrival of a nearby competitor. In the months after entry, the price of regular unleaded gasoline declines by 2.7 cents per gallon in our preferred specification.¹⁷ Economically, this effect represents roughly 7% of the average station gross markup and about \$44,000 in annual revenue per station.¹⁸ The estimates are precisely measured and persist over time, consistent with entry inducing a rapid transition to a new, lower-price equilibrium. In contrast, Figure 2b shows precisely estimated null effects following competitor exit, reinforcing the asymmetric price response between events.

The preceding results focus on the level of incumbent prices. A natural extension is to examine relative pricing by assessing whether entering and exiting stations set prices systematically above or below nearby incumbents. As descriptive evidence, we calculate the daily percentile rank of each entering or exiting station's price for regular gasoline relative to incumbent stations within a 1-mile radius. On average, entering stations price somewhat more aggressively than incumbents, operating at the 42nd percentile of the local price distribution. Prices at exiting stations are close to the median, at the 48th percentile on average leading up to their closure. Consistent with the persistence of incumbent price responses, we find little evidence that the relative pricing of entering or exiting stations changes after entry or in the period leading up to exit. Event-month averages by group similarly show no clear trend in relative price ranks.

To further connect our findings to the literature on station competition, we examine heterogeneity by the geographic proximity of the entering or exiting station. Gasoline retail markets are geographically differentiated, with nearby stations competing more intensely than distant rivals (Houde 2012; Chandra and Tappata 2011; Eckert and West 2005). If entrants impose meaning-

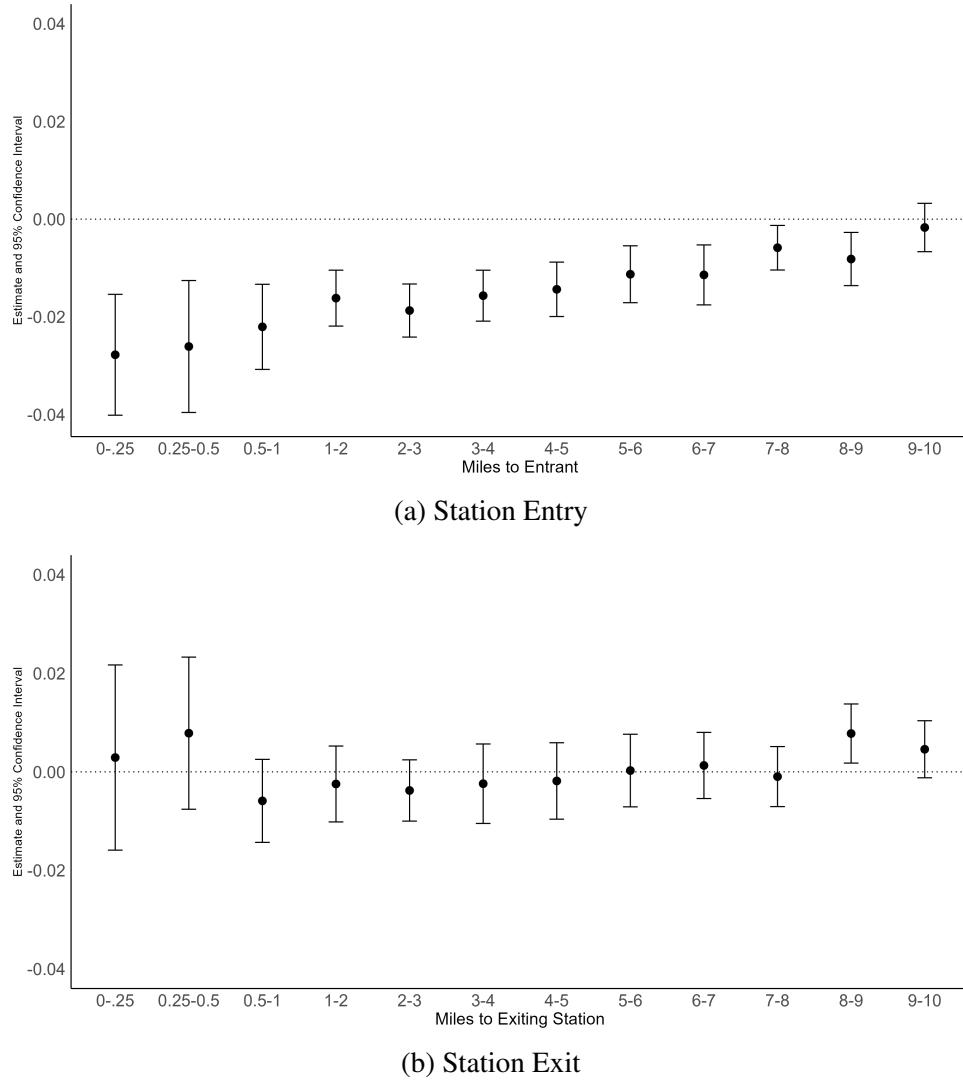
¹⁷For completeness, we replicate the analysis for the gasoline prices of mid-grade and premium gasoline. Building on Yatchew and No (2001) and Hastings and Shapiro (2013) that find evidence of substitution between grades of gasoline, we test whether entry or exit affect pricing by grade. We present the estimates for the effect on incumbent pricing for all three grades for entry in Appendix Figure A2a and exit in Appendix Figure A2b. We find qualitatively consistent results across the three grades. Although the point estimates for the price decrease of regular grade gasoline is the steepest, we cannot statistically reject that all three blends have the same coefficient. As was the case for regular gasoline, exit events are not associated with statistically distinguishable changes to incumbent pricing for either of the other two blends.

¹⁸Average annual gallons sold per station during 2014–2018, calculated from the CEC A15 survey, are multiplied by the estimated short-run entry effect to obtain the implied revenue impact.

ful competitive pressure on incumbents, the price response should be largest following very close entry and attenuate with distance, all else equal.

We examine how effects vary with geographic proximity using the specification in equation (2), estimating entry and exit separately. We estimate the models separately for predefined distance buckets, defining treatment as the month of the first entry or exit by a station within each bucket's range. As in the baseline specification, we measure competition using the straight-line distance between incumbent and entering stations. Although this abstracts from road networks and commuting patterns used in similar work ([Houde 2012](#); [Davis et al. 2022](#)), it allows for a consistent statewide implementation. Figures 3a and 3b report the resulting point estimates and standard errors. As before, stations that do not experience entry or exit serve as controls.

Figure 3: Effect of Station Entry and Exit on Incumbent Pricing, by Distance



Notes: Graph plots the coefficient estimates and 95% confidence intervals for the effect of station entry and exit by distance from incumbent station on incumbent pricing of regular unleaded gasoline. Coefficients are estimated by running separate regressions by distance bucket. Regressions are estimated using the two-stage DID estimator in [Gardner et al. \(2024\)](#). Standard errors are clustered at the city level. Data are sourced from the Oil Price Information Service (OPIS) for all stations in California for 2014-2018.

Entry effects decline with distance from the incumbent, as expected, with the largest price responses observed for events within a one-quarter mile. The magnitude of the effect attenuates steadily as distance increases and becomes economically negligible beyond seven miles. This pattern is consistent with the localized nature of retail gasoline competition documented in prior work. In contrast, we again estimate null effects for exit across distance bins, reinforcing the

asymmetric response between entry and exit events.

Identification of causal estimates in our setting relies on the SUTVA assumption. Our distance results show that this assumption is valid for most distances with limited spatial spillovers of entry. Our main results are robust to exclusion of control stations which do not have entry within 1 mile, but do experience an entry within the 7 mile cutoff for strong significant effects, eliminating potential spillovers.¹⁹

5 Asymmetry of Entry and Exit Effects

We find that incumbent prices decline immediately following entry, but find little evidence that exit leads to price increases. This raises a natural question: does the asymmetry reflect something fundamental about entry and exit, or does it instead arise from differences in the types of stations that enter and exit or in the locations where entry and exit occur?

Asymmetry can arise in competitive models in which entering and exiting firms differ. For instance, in a standard logit model with single-product firms setting prices for differentiated products, the first-order condition for profit maximization is:

$$p_i = c_i + \frac{1}{\alpha(1 - s_i)} \quad (5)$$

where c_i is the marginal cost for firm i , s_i is the market share of firm i , and α is the disutility of the price of the good.²⁰ The potential impact of entry or exit on the price of incumbent i depends on how much the event will change incumbent i 's market share. If entry or exit has little impact on demand for the incumbents' goods (e.g., if the entering or exiting firm sells a poor substitute or the market is highly competitive), we would expect smaller impacts on incumbent pricing than if the event caused the market share of incumbents to change significantly (e.g., if the entering or

¹⁹Appendix Figure A3 reports the coefficients and shows that estimated results are even stronger at the time of entry, persist at the increased level, and still support the parallel trends assumption. Confidence intervals remain similar in magnitude despite the reduction in sample size.

²⁰We formally derive the profit-maximization condition from the standard logit model in Appendix C.

exiting firm offers a very attractive product or there are few incumbents). This framework provides intuition for why entry effects may attenuate with distance and with the number of incumbent competitors (e.g., [Bresnahan and Reiss \(1991\)](#), [Tappata and Yan \(2017\)](#)).

This framework suggests that the impact of entry or exit will depend both on the competitiveness of the affected market and on the characteristics of the entering or exiting firm. In our setting, we explore two plausible avenues through which the asymmetry might arise. First, as we note in Table 1, station entry and exit occur in observably different locations. Entry is more likely to occur in locations with fewer competitors, further from highways, and with lower electric vehicle penetration.²¹ If entry disproportionately occurs in less competitive markets, or in locations where competitive spillovers are larger, we might expect to see a larger pricing response by incumbents in response to entry rather than exit.²²

Second, entering and exiting stations differ along observable dimensions that may affect their impact on incumbent stations. Returning to Table 1, entering stations are less likely to be branded and more likely to be hypermarkets than exiting stations.²³ Entering and exiting stations may also differ along unobservable dimensions. For example, [Asplund and Nocke \(2006\)](#) note that in models of endogenous exit with heterogeneous costs or productivity, higher-cost or lower-productivity stations are more likely to exit. In our setting, this might manifest if older stations with less attractive amenities or fewer pumps are more likely to exit. If such stations exert weaker competitive pressure on neighboring firms, the first-order condition implies that their exit may have a smaller price effect than the entry of a new, more attractive competitor.

²¹A long literature (e.g., [Manuszak and Moul \(2008\)](#), [Seim \(2006\)](#)) highlights the geographic element of endogenous entry decisions, in which firms selectively enter more profitable markets.

²²Notably, there is not a clear consensus in the literature on the effect of the number of competitors on gasoline prices. Canonical work by [Bresnahan and Reiss \(1991\)](#) on competition in homogeneous goods markets suggests that entry into markets with fewer competitors results in larger competitive effects that dissipate as competitors increase. However, [Armstrong and Vickers \(2022\)](#) and [Barron et al. \(2004\)](#) show that this result can reverse in search models with price dispersion depending on the search costs involved.

²³Endogenous entry by low-cost firms is consistent with our findings in Section 4 that entering stations tend to set lower prices relative to nearby incumbents than exiting stations.

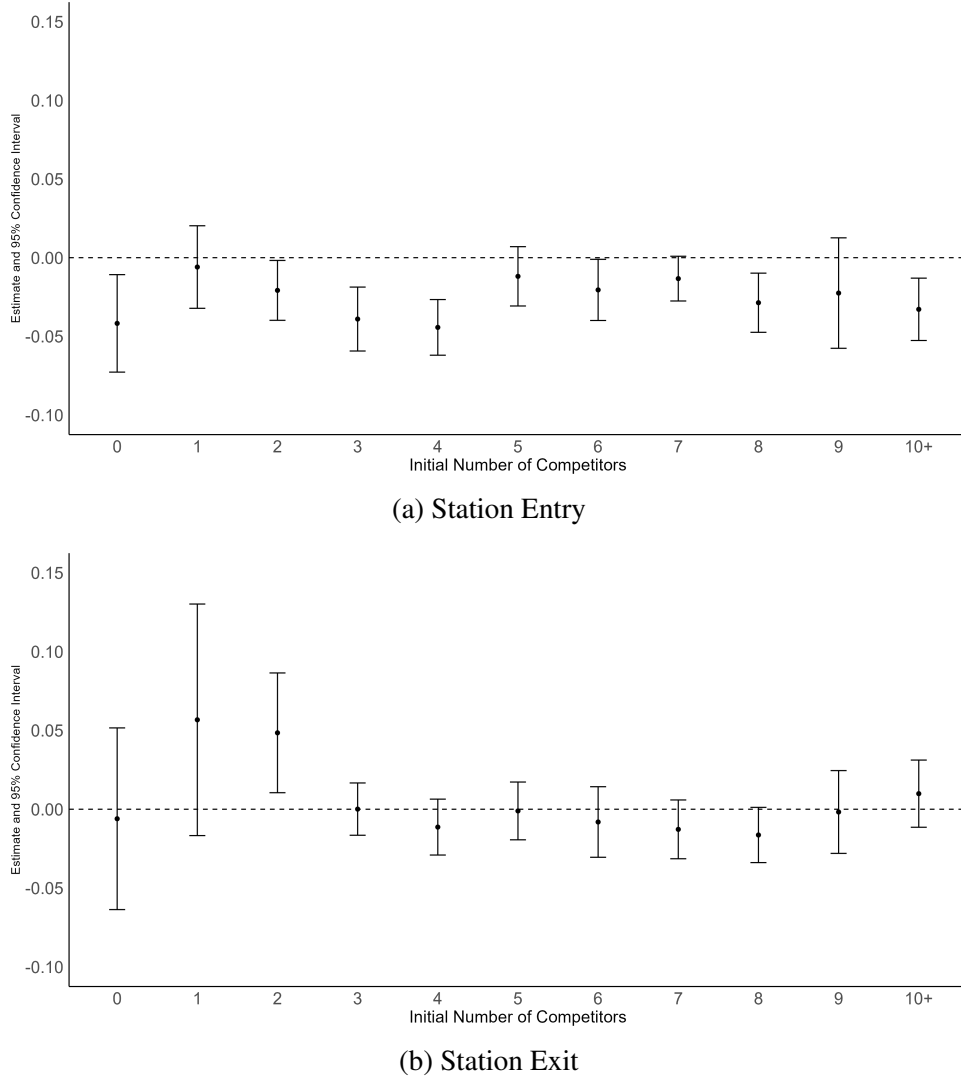
5.1 Compositional differences in the location or type of entering and exiting station

Guided by the framework above, we explore each channel by estimating the specification from equation (2) for various subsamples of stations. To test whether observable locational differences explain the asymmetry, we subsample based on the number of nearby competitors, the distance to highways, and local electric vehicle penetration.

Figure 4 plots the coefficients for entry (in panel (a)) and exit (in panel (b)) by the number of competitors faced by the incumbent station at the beginning of the sample. We find that incumbent stations facing zero, two, three, or four competitors within one mile lower prices upon nearby entry. These cases make up the majority of the entry events we observe. Roughly two-thirds of the incumbents affected by entry face fewer than five competitors.²⁴ We find little evidence that entry affects the prices of incumbents facing more than four competitors, similar to Tappata and Yan (2017) who find little incremental impact on fuel prices once six competitors are present. For exit events, we do estimate positive effects for incumbents facing very few competitors. Notably, point estimates suggest that incumbent stations with one or two competitors raise prices by roughly five cents per gallon when a competitor exits, although only the estimate for the latter is statistically significant. Yet these cases account for a relatively small fraction (15%) of the roughly 350 exit events in the sample. We find robust null effects of exit for the remaining 85% of cases. As a point of comparison, incumbents with one or two competitors comprised a larger fraction (roughly 25%) of stations affected by entry. Consistent with the framework above, exit effects appear only when the exiting station plausibly represents a meaningful competitive constraint.

²⁴We show the full distribution of initial market size for entry and exit events in Appendix Figure A4.

Figure 4: Effect of Station Entry and Exit on Incumbent Pricing, by Initial Market Size



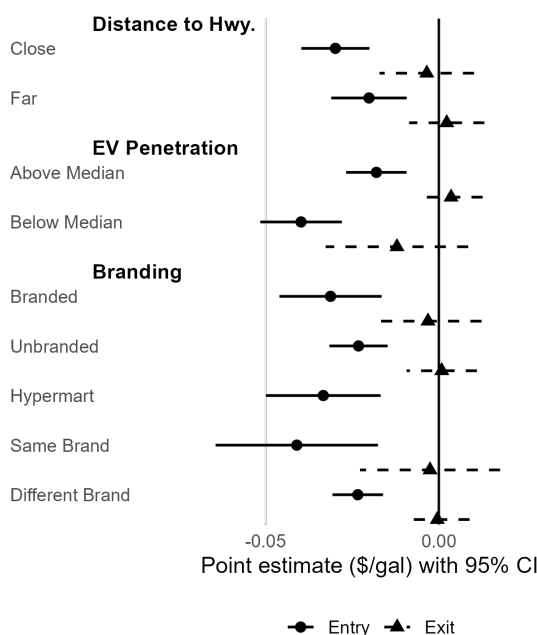
Notes: Graph plots the coefficient estimates and 95% confidence intervals for the effect of station entry and exit within 1 mile on incumbent pricing of regular unleaded gasoline. Coefficients are estimated by running separate regressions by the number of initial competitors faced. Regressions are estimated using the two-stage DID estimator in [Gardner et al. \(2024\)](#). Standard errors are clustered at the city level. Data are sourced from the Oil Price Information Service (OPIS) for all stations in California for 2014-2018.

We further explore compositional differences by estimating equation (2) separately by high-way proximity and electric vehicle penetration. Figure 5 presents estimates for the resulting sub-samples.²⁵ In each case, entry estimates are shown in blue and exit estimates in red. Comparing the effects of entry, we find modestly larger impacts near to highways (less than .25 miles) and in

²⁵We present event-study versions for the results in Figure 5 in Appendix Figures A5–A8.

areas that have experienced slower-than-average electric vehicle adoption. Notably, markets with fewer EVs exhibit deeper long-run entry effects, possibly driven by differences in the type of entering station or by secondary entry further depressing prices. Yet across all four subsamples, we find null effects of exit on incumbent pricing. Although there are statistically significant differences in where entry and exit occur, we find little evidence that these compositional differences explain the asymmetric effects observed in the full sample.²⁶

Figure 5: Heterogeneity of Effects by Observable Characteristics



Notes: Graph plots the coefficient estimates and 95% confidence intervals for the effect of station entry and exit within 1 mile on incumbent pricing of regular unleaded gasoline. Coefficients are estimated by running separate regressions based on highway proximity, electric vehicle penetration, station type, and branding. Regressions are estimated using the two-stage DID estimator in [Gardner et al. \(2024\)](#). Standard errors are clustered at the city level. Data are sourced from the Oil Price Information Service (OPIS) for all stations in California for 2014-2018.

Figure 5 also presents estimates by the type and branding of the entering or exiting station. Consistent with the previous results, we find little evidence that compositional differences in the types of stations that enter or exit explain the asymmetric effects. Incumbent prices fall by roughly 2–3 cents per gallon following entry, regardless of whether the entrant is branded, unbranded, a

²⁶In fact, compositional differences would tend to attenuate, rather than amplify, the observed asymmetry, as entering stations are more likely to be located *farther* from highways than exiting stations.

hypermart, or shares the incumbent’s brand. In contrast, exits do not have a statistically significant effect on incumbent pricing.²⁷

5.2 Heterogeneity in Unobservables

Finally, we explore whether selection plays a role in explaining the null effects of exit we observe. As noted above, if marginal stations exert little competitive pressure on nearby stations prior to exit, either because they are not cost competitive or because they offer a less attractive “product”, their exit might not impact prices at neighboring stations. To test this, we construct a proxy for the “unobserved competitiveness” of exiting stations by using the frequency with which OPIS reports prices prior to exiting.

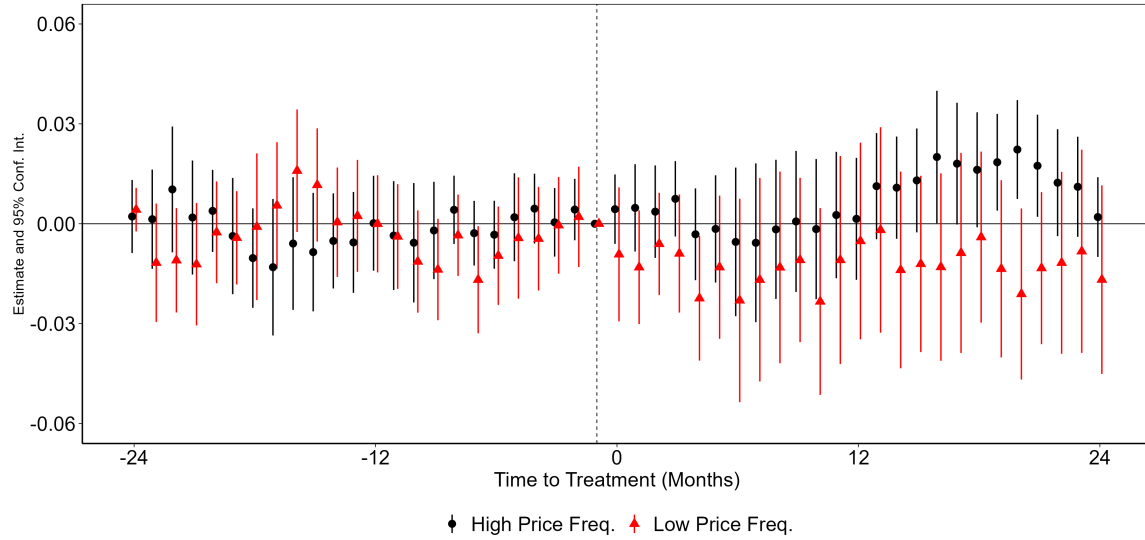
OPIS collects price data in several ways, two of which, card swipes and consumer-reported prices through GasBuddy, require a customer to purchase fuel at a station or observe prices there. Because of this, reporting frequency may proxy for station traffic and unobserved competitiveness. As evidence, the OPIS data reports a price on 97% of the days for the median incumbent station. In sharp contrast, the median exiting station has a price observation on roughly 50% days prior to exit.²⁸

To examine the role of selection, we first split exit events at the median of price reporting frequency, plotting event-study coefficients in Figure 6. Comparing the estimates for high-frequency exiting stations (in black) and low frequency exiting stations (in red), we find suggestive evidence that high-frequency exits have larger price impacts post exit. The point estimates for high-frequency station exits are uniformly higher than those for low-frequency stations, albeit in many cases, not statistically distinguishable.

²⁷We characterize same branding using the store brand of stations. We alternatively use the gas branding, which can be different, with similar results in Appendix Figure A9. We omit the noisy hypermart exit coefficient as only eight exit our sample.

²⁸We show the CDF for the frequency of price reporting for incumbent and exiting station in Appendix Figure A10.

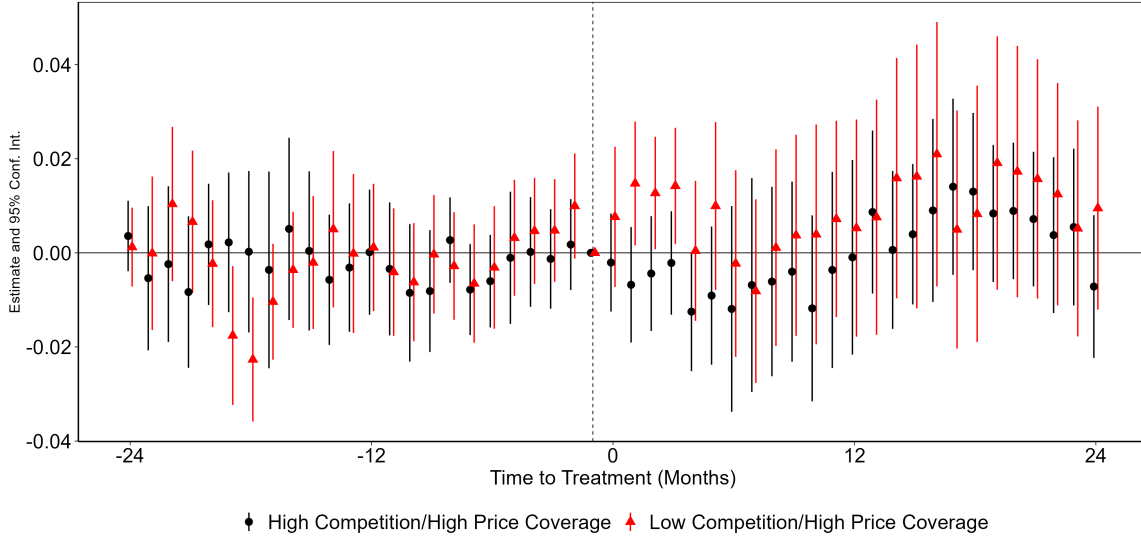
Figure 6: Effect of Station Exit, by Reporting Frequency



Notes: Graph plots the coefficient estimates and 95% confidence intervals for the effect of station exit within 1 mile on incumbent pricing by exiting stations with above and below median price reporting frequency. Regressions are estimated using the two-stage DID estimator in [Gardner et al. \(2024\)](#). Standard errors are clustered at the city level. Data are sourced from the Oil Price Information Service (OPIS) for all stations in California for 2014-2018.

We next examine whether the price impact of exit for the high-frequency stations varies across markets with few versus many competitors. Guided by the previous framework, highly competitive stations in markets with fewer competitors plausibly impact the market share of competitors and thus impact prices. Figure 7 plots the event-study estimates for high-frequency reporting stations with above median market competitors (i.e., four or more) in black and below median market competitors in red. Notably, when highly competitive stations depart from areas with a high number of competitors, we find little impact (as expected) on incumbent pricing. However, when a highly competitive station exits a market with few incumbent stations, we now estimate an immediate and statistically significant increase in prices that are similar in magnitude to the estimated entry effect. Taken together, these results suggest that the null effect of exit reflects a combination of selection in which stations exit and the number of competitors in markets where exits occur.

Figure 7: Effect of High Price Coverage Station Exits, by Market Size



Notes: Graph plots the coefficient estimates and 95% confidence intervals for the effect of the exit of a high-frequency reporting station within 1 mile on incumbent pricing. Results are separated by stations with above and below median nearby competitors prior to exit. Regressions are estimated using the two-stage DID estimator in [Gardner et al. \(2024\)](#). Standard errors are clustered at the city level. Data are sourced from the Oil Price Information Service (OPIS) for all stations in California for 2014-2018.

6 Policy Implications and Discussion

Roughly one quarter of total U.S. carbon emissions come from on-road combustion, making the retail gasoline market central to policies aimed at decarbonizing light-duty transportation. Our estimates of the price effects of station entry and exit have three policy-relevant implications. First, they speak to how zoning restrictions and environmental regulations that impede station entry or encourage exit, such as local bans on new stations or stringent underground storage tank requirements, may translate into higher retail gasoline prices.²⁹ Second, they inform the extent to which such price changes, as well as changes in gasoline availability, may influence vehicle purchase decisions during the energy transition. Third, they clarify how regulations affecting gasoline stations may interact with public support for transportation decarbonization policies, given the salience of gasoline prices to voters.

²⁹[Gyourko et al. \(2021\)](#) provide additional evidence that land-use regulation has become increasingly restrictive over time.

Our work informs the potential price impacts of zoning restrictions or environmental hurdles that either limit entry or encourage exit. Our results complement the vast literature that explores the effects of policies that directly impact gasoline prices, including studies on the pass-through of fuel taxes (e.g., [Marion and Muehlegger \(2011\)](#)), renewable fuel credit prices (e.g., [Knittel et al. \(2017\)](#), [Lade and Bushnell \(2019\)](#)), and studies of the price effects of boutique fuel blends (e.g., [Walls and Rusco \(2007\)](#), [Brown et al. \(2008\)](#)). Local jurisdictions can exercise substantial discretion through the permitting and zoning process increasingly using this authority to place restrictions on station construction by placing prohibitions (Sonoma County), significant limitations (Denver, CO), or moratoria (East Hartford, CT). Likewise, although federal rules have required new or replaced underground tanks to be double-walled since 2016, states possess the discretion to place additional requirements on existing stations, such as California’s requirement to close any single-walled tanks by December 31, 2025.

Although we do not estimate the effect of regulation on entry or exit directly, our estimates provide a benchmark for the price impacts that would arise if these policies reduced marginal entry or induced marginal exit. These impacts are relevant for welfare and environmental policy. All else equal, higher prices paid by consumers would tend to increase profits for owners of legacy infrastructure. In parts of California outside of Los Angeles and San Francisco where prices exceed social marginal costs (see e.g., [Borenstein and Bushnell \(2022\)](#)), price increases would lower overall welfare. Constraints on entry or induced exit may also impose costs on consumers through reduced convenience as drivers may need to travel farther to refuel or face longer wait times at remaining stations, implying further welfare losses. Yet, in regions where retail gasoline prices are below social marginal cost, price impacts may reduce the gap between retail prices and social marginal cost. Finally, these burdens might not be equitably shared. A long literature establishes the disproportionate environmental burden borne by low socioeconomic status and less politically connected neighborhoods (see, e.g., [Banzhaf et al. \(2019\)](#)). Here, if local zoning restrictions or environmental policy induce exit or prevent station entry, the burdens and welfare impacts might be more pronounced in affluent, politically-connected communities.

Second, station entry or exit has implications for vehicle purchase decisions during the transition towards electric vehicles. Changes in the number of stations can affect both gasoline prices and the convenience of gasoline access, which may influence durable vehicle choices. Our estimates suggest that the short-run price effects of entry or exit are likely to have modest impacts on vehicle purchase decisions at the scales observed in our setting. A 2–3 cent per gallon price reduction associated with a single entrant implies annual savings of approximately \$8–\$12 for a gasoline vehicle driven 12,000 miles per year at 30 miles per gallon. These magnitudes are small relative to gasoline price changes shown in prior work to affect vehicle choice (e.g., [Busse et al. \(2013\)](#)) and the impact of gasoline prices on electric vehicle purchases (e.g., [Bushnell et al. \(2022\)](#)). Yet, it is important to note that our identification strategy best captures short-run price adjustments. Our results may not naturally extrapolate to the effects of a long-run shift away from gasoline or the effects of legacy station exit driven by environmental restrictions, rather than endogenous selection. Whether continued exit of gasoline stations leads to larger price impacts or sufficiently reduces convenience to impact vehicle purchase decisions remains an open question.

Finally, our results inform the political economy of transportation decarbonization by clarifying the extent to which station regulations plausibly contribute to high and politically salient gasoline prices. Gasoline prices are a prominent concern for both politicians and voters and correlated with a decline in popular support for gasoline taxes ([Epstein and Muehlegger \(2024\)](#)) and restrictions to limit sales of conventional vehicles ([Hernandez Carballo and Sisco \(2024\)](#)). This is particularly relevant for California where drivers face some of the highest gasoline prices in the nation. The state uses both taxes and vehicle restrictions as tools to speed the transition towards electric vehicles and policy is shaped in part by popular ballot measures. Our results suggest that environmental and zoning regulations are unlikely to explain, in isolation, California’s high gasoline prices. California is roughly in the middle of the pack in terms of gasoline stations per square mile³⁰ with higher station density than Texas, but lower station density than other large states like Illinois and New York. Nor does our analysis provide a clear explanation for California’s “mystery

³⁰<https://coastpay.com/ev-charging-stations-vs-gas-stations-comparing-density-in-the-u-s/>

gasoline surcharge”³¹ which represents 30 - 65 cents per gallon of the premium paid by California drivers, relative to drivers in other states since 2015. Station counts in California did not change sharply at the start of the surcharge³² and the magnitude of the surcharge is significantly greater than the estimated effect of an entrant (roughly 2.7 cents per gallon).

7 Conclusion

Using daily price data and the precise geographic location for gas stations in California, we estimate the effect of entry and exit of a nearby gas station on the pricing for incumbent stations. The use of high-frequency data and the ability to restrict identification to within-station variation in the difference-in-differences and event-study approaches accounts for the endogenous entry and exit decisions of firms. We find that nearby entry is associated with an immediate and statistically significant 2.7-cent decrease in the price at incumbent stations, reflecting a 7% decrease in average retail markups. In contrast, we estimate that a nearby exit has a precise null effect on incumbent pricing from the exit of a nearby firm.

Two features help to explain the asymmetry between the effects of entry and exit in our setting. First, entry tends to occur in more concentrated markets than exit, where the impact of a change in competitors is greater. Second, we find suggestive evidence that the null effects of exit are driven partially by the relative competitiveness of entering and exiting stations. Since the OPIS data relies (at least partially) on transaction data and crowd-sourced customer reporting, the frequency with which prices are observed serves as a potential proxy for the unobservable attributes reflecting the competitiveness or attractiveness of a station. Isolating the subset of exiting stations with frequent price reports, we find positive price effects on incumbent pricing similar in magnitude to the entry effect. This finding suggests important potential implications for the ongoing transition

³¹For background discussion, see Borenstein’s testimony before the California Assembly Select Committee on Gasoline Supply and Pricing. <https://www.assembly.ca.gov/sites/assembly.ca.gov/files/borensteingasolinetestimony220630.pdf>

³²https://afdc.energy.gov/files/u/data/data_source/10333/10333_gasoline_stations_year.xlsx

away from petroleum-based transportation fuels. If declining demand leads to exit of the least competitive stations, the effects on gasoline and diesel prices at the remaining stations is likely to be modest. But, if station exit is driven by factors orthogonal to station competitiveness (e.g., local zoning or environmental restrictions) rather than endogenous selection, exit may lead to higher prices. Informing how a long-run shift away from gasoline impacts station exit or incumbent pricing remains an important direction for future research.

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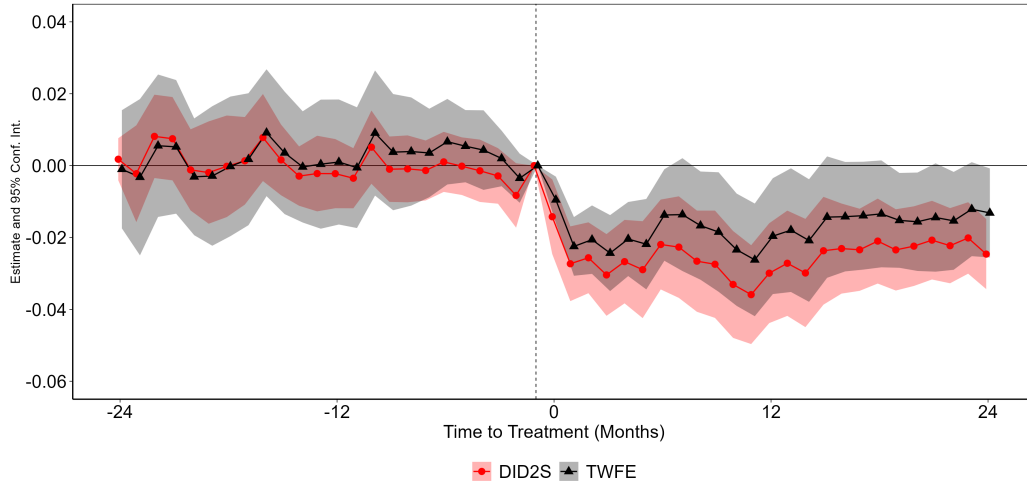
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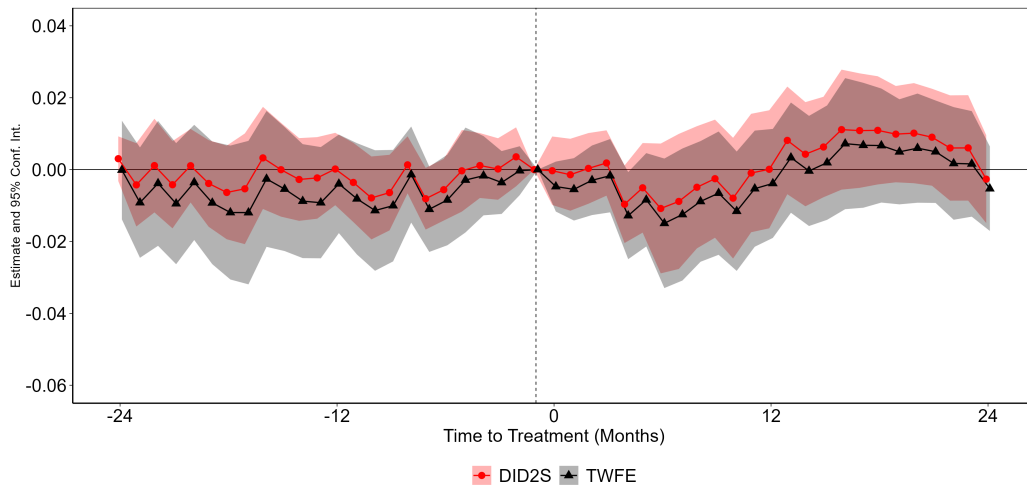
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A Appendix: Figures

Appendix Figure A1: Effect of Station Entry and Exit Within 1 Mi. on Incumbent Pricing



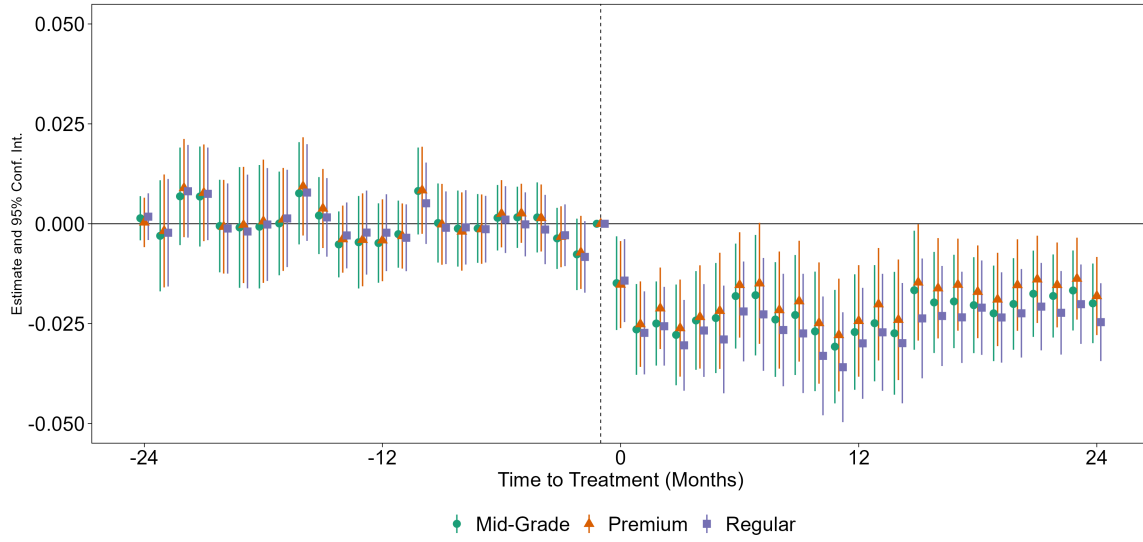
(a) Station Entry



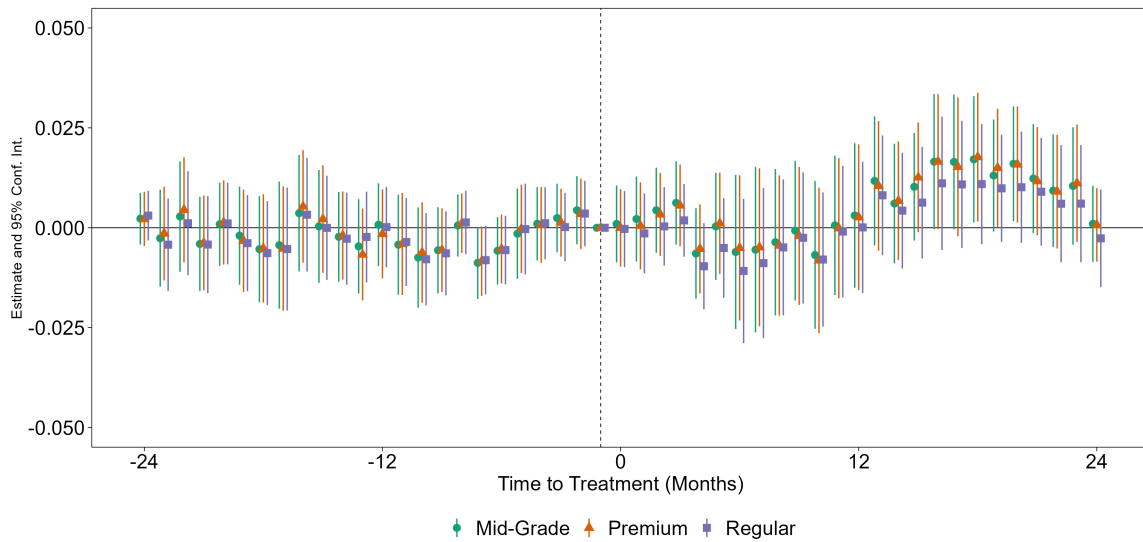
(b) Station Exit

Notes: Graph plots the coefficient estimates and 95% confidence intervals for the effect of station entry (in panel (a)) and station exit (in panel (b)) within 1 mile of an incumbent station on incumbent pricing of regular unleaded gasoline. Results from the canonical TWFE estimator are shown in black. Results from the two-stage DID estimator from [Gardner et al. \(2024\)](#) are shown in red. Standard errors are clustered at the city level. Data are sourced from the Oil Price Information Service (OPIS) for all stations in California for 2014-2018.

Appendix Figure A2: Effect of Station Entry and Exit on Incumbent Pricing, by Blend



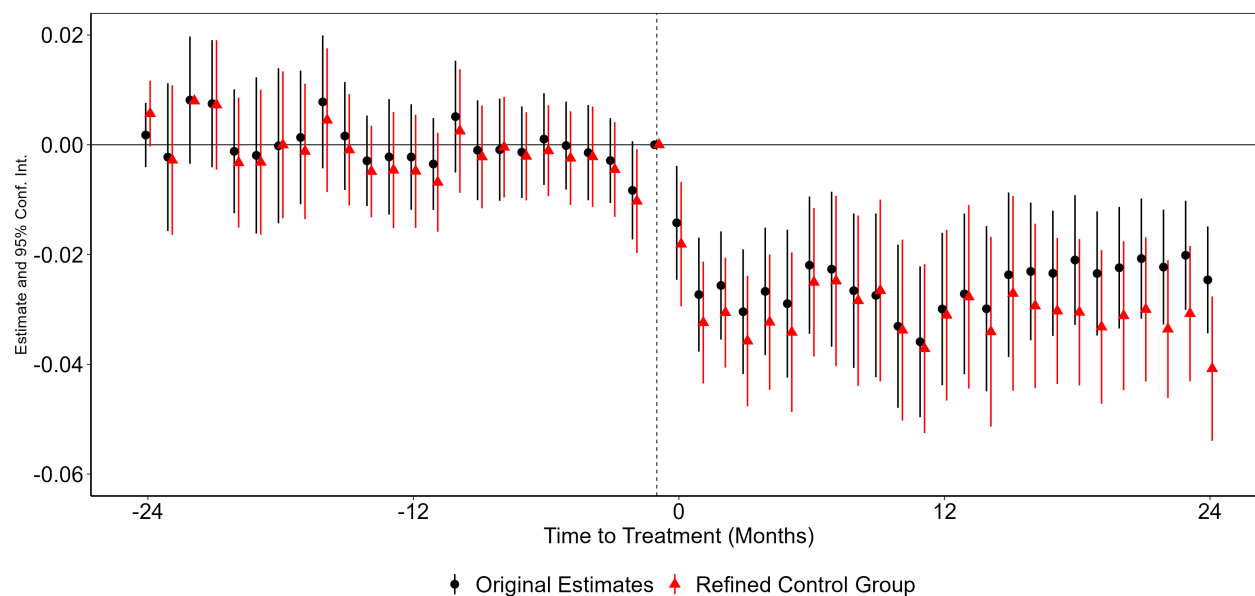
(a) Station Entry



(b) Station Exit

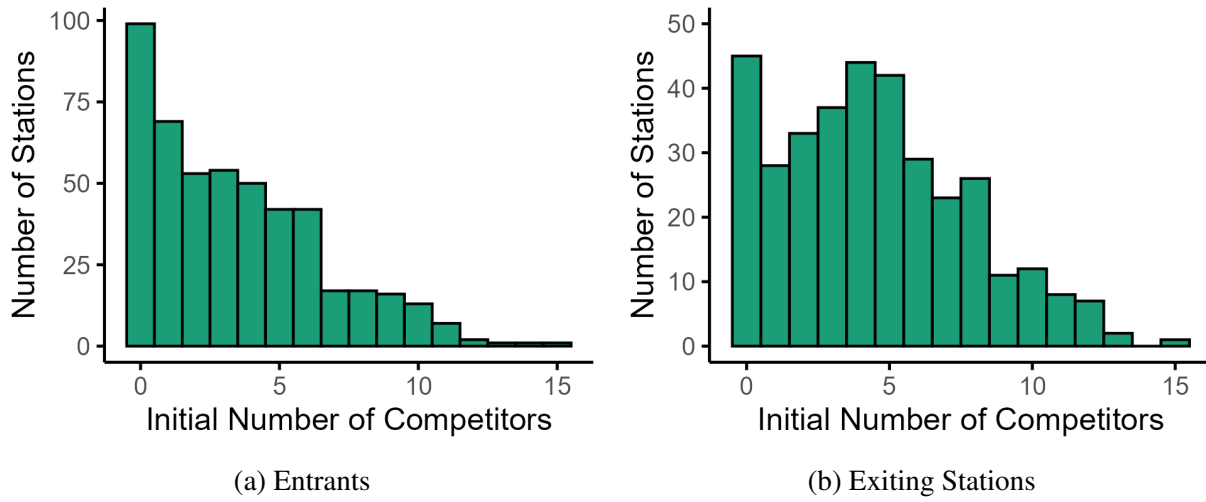
Notes: Graph plots the coefficient estimates and 95% confidence intervals for the effect of station entry (in panel (a)) and station exit (in panel (b)) within 1 mile of an incumbent station on incumbent pricing by gasoline blend type. Regressions are estimated using the two-stage DID estimator in [Gardner et al. \(2024\)](#). Standard errors are clustered at the city level. Data are sourced from the Oil Price Information Service (OPIS) for all stations in California for 2014-2018.

Appendix Figure A3: Effect of Station Entry on Incumbent Pricing excl. Intermediate Distances



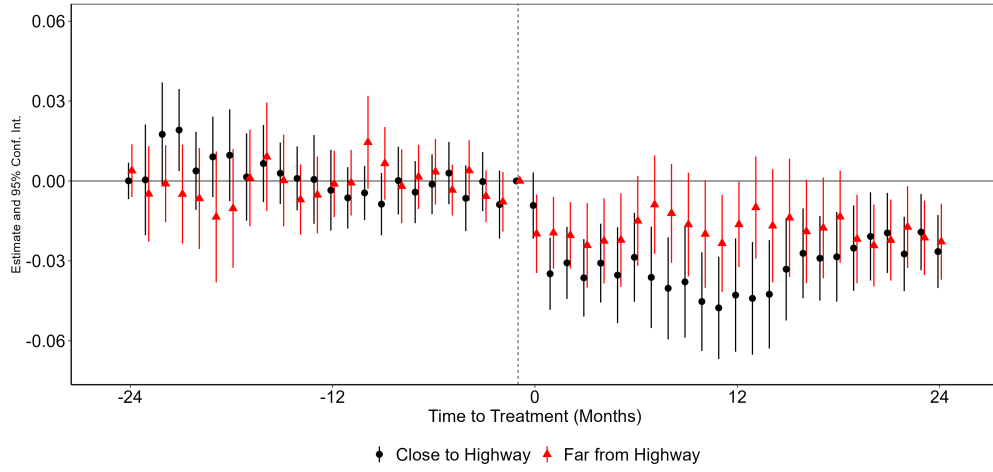
Notes: Graph plots the coefficient estimates and 95% confidence intervals for the effect of station entry within 1 mile of an incumbent station on incumbent pricing are shown. The refined control group removes control stations that experienced an entry between 1-7 miles during the sample. Regressions are estimated using the two-stage DID estimator in [Gardner et al. \(2024\)](#). Standard errors are clustered at the city level. Data are sourced from the Oil Price Information Service (OPIS) for all stations in California for 2014-2018.

Appendix Figure A4: Entries and Exits, by Initial Market Size

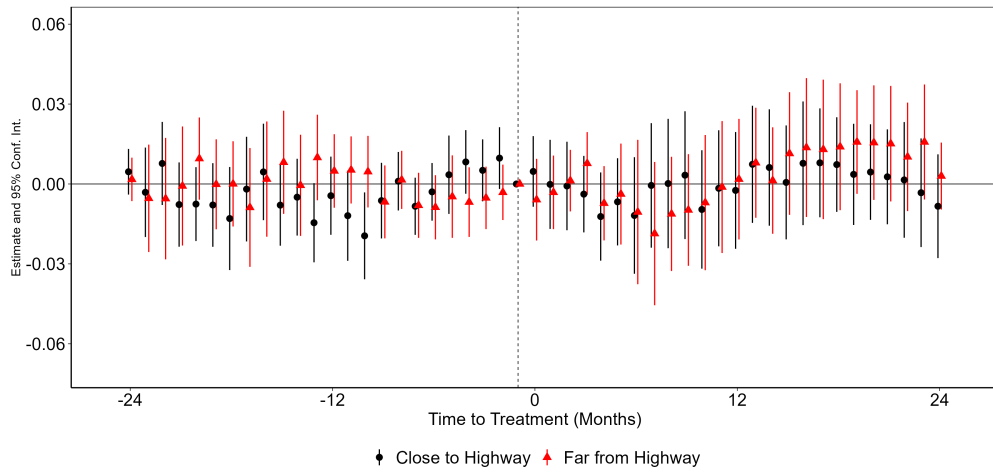


Note: Graph plots the initial market size for entering and exiting stations. Data are sourced from the Oil Price Information Service (OPIS) for all stations in California for 2014-2018.

Appendix Figure A5: Effect of Station Entry and Exit, by Distance to Highway



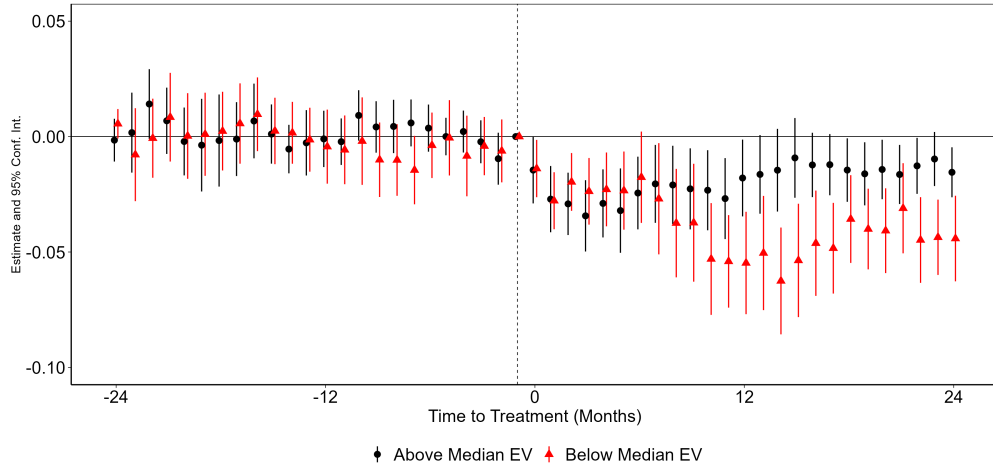
(a) Station Entry



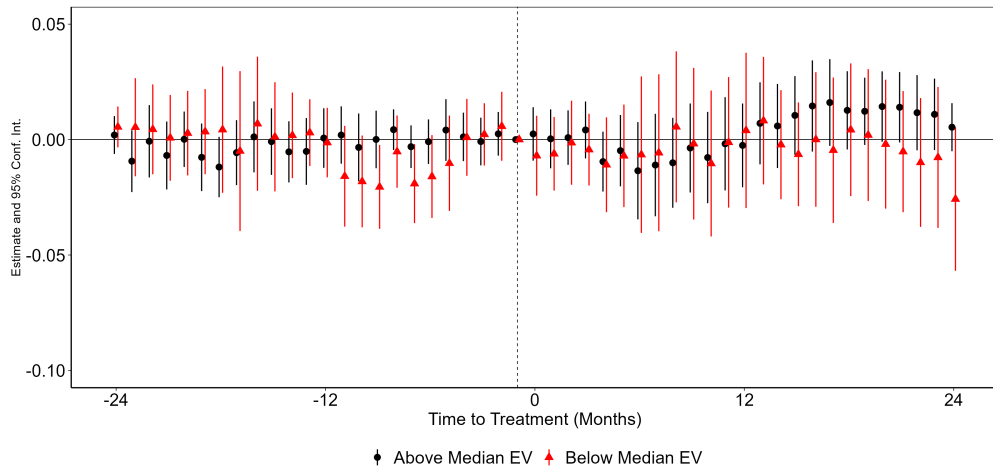
(b) Station Exit

Notes: Graph plots the coefficient estimates and 95% confidence intervals for the effect of station entry (in panel (a)) and station exit (in panel (b)) within 1 mile on incumbent pricing of regular unleaded gasoline. Close to highway is defined as being less than one quarter mile to a highway or interstate. Regressions are estimated using the two-stage DID estimator in [Gardner et al. \(2024\)](#). Standard errors are clustered at the city level. Data are sourced from the Oil Price Information Service (OPIS) for all stations in California for 2014-2018.

Appendix Figure A6: Effect of Station Entry and Exit, by EV Penetration



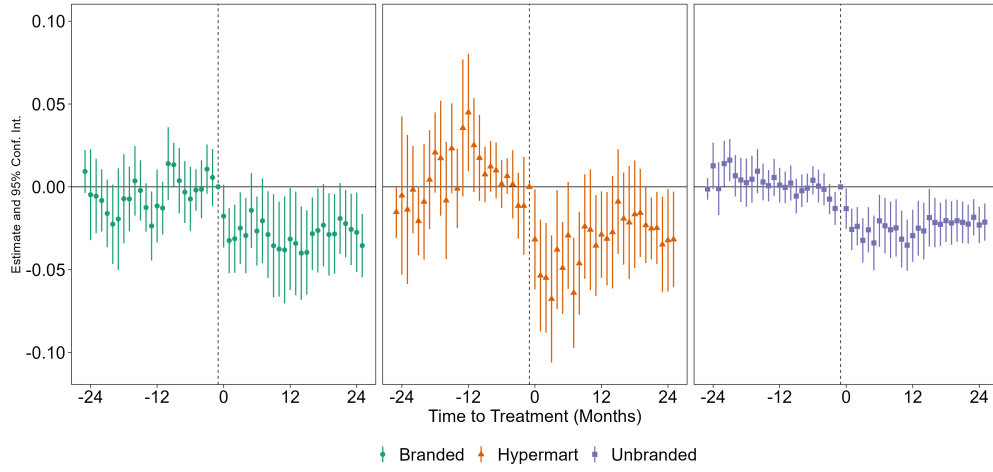
(a) Station Entry



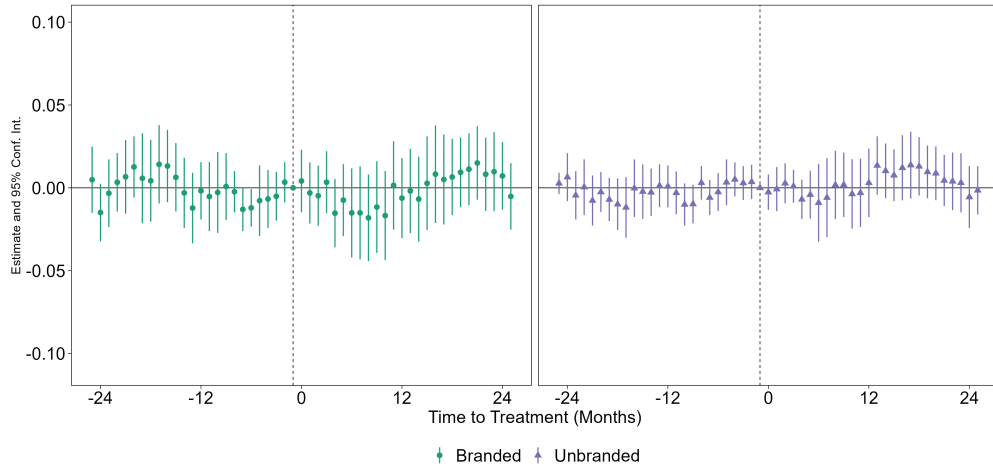
(b) Station Exit

Notes: Graph plots the coefficient estimates and 95% confidence intervals for the effect of station entry (in panel (a)) and station exit (in panel (b)) within 1 mile by level of local EV penetration on incumbent pricing of regular unleaded gasoline. Regressions are estimated using the two-stage DID estimator in [Gardner et al. \(2024\)](#). Standard errors are clustered at the city level. Data are sourced from the Oil Price Information Service (OPIS) for all stations in California for 2014-2018.

Appendix Figure A7: Effect of Station Entry and Exit, by Station Category



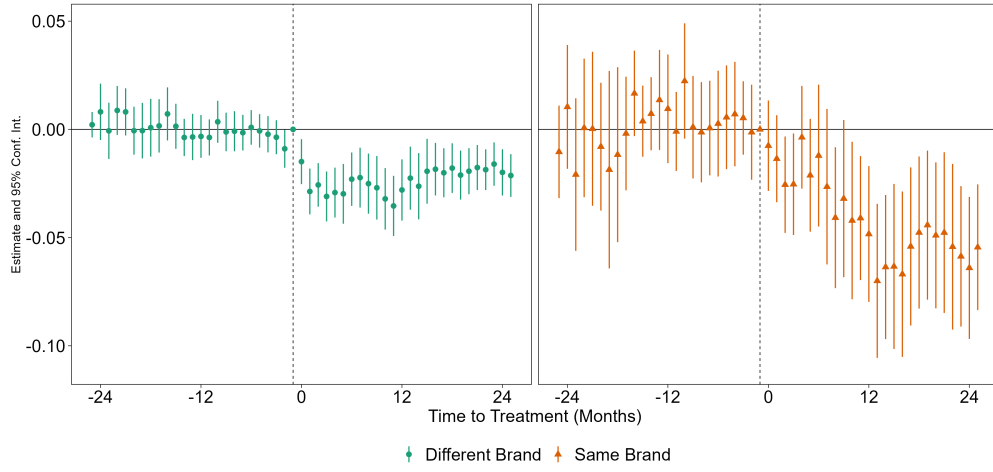
(a) Station Entry



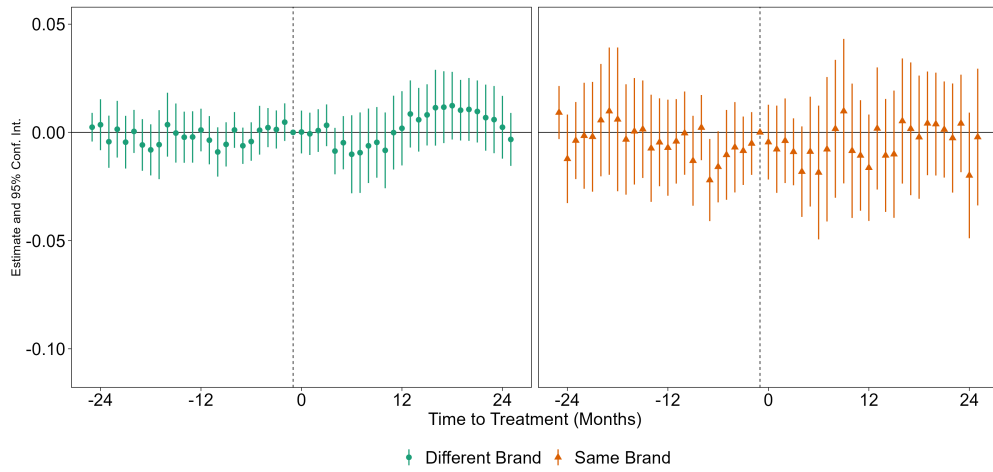
(b) Station Exit

Notes: Graph plots the coefficient estimates and 95% confidence intervals for the effect of station entry (in panel (a)) and exit (in panel (b)) within 1 mile on incumbent pricing of regular unleaded gasoline by station brand type. Hypermarts are omitted from panel (b) as there are too few hypermarts that exit the sample. Regressions are estimated using the two-stage DID estimator in [Gardner et al. \(2024\)](#). Standard errors are clustered at the city level. Data are sourced from the Oil Price Information Service (OPIS) for all stations in California for 2014-2018.

Appendix Figure A8: Effect of Station Entry and Exit, by Station Branding



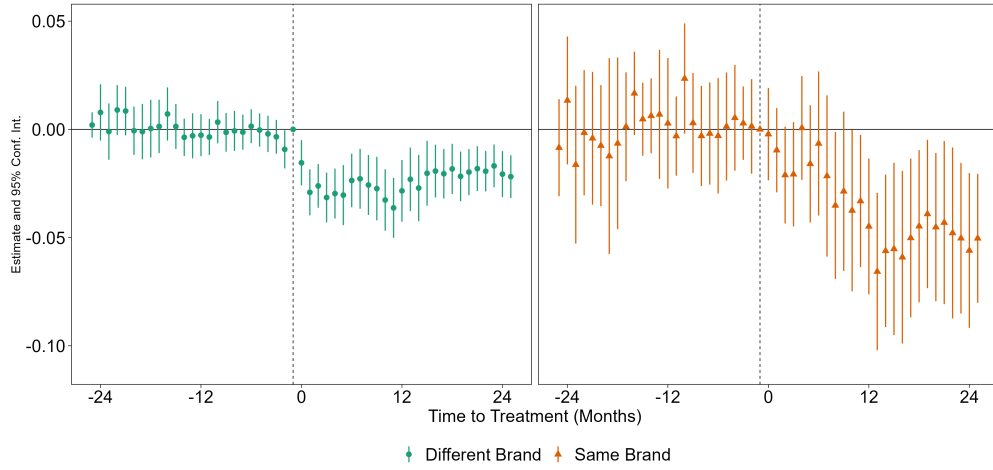
(a) Station Entry



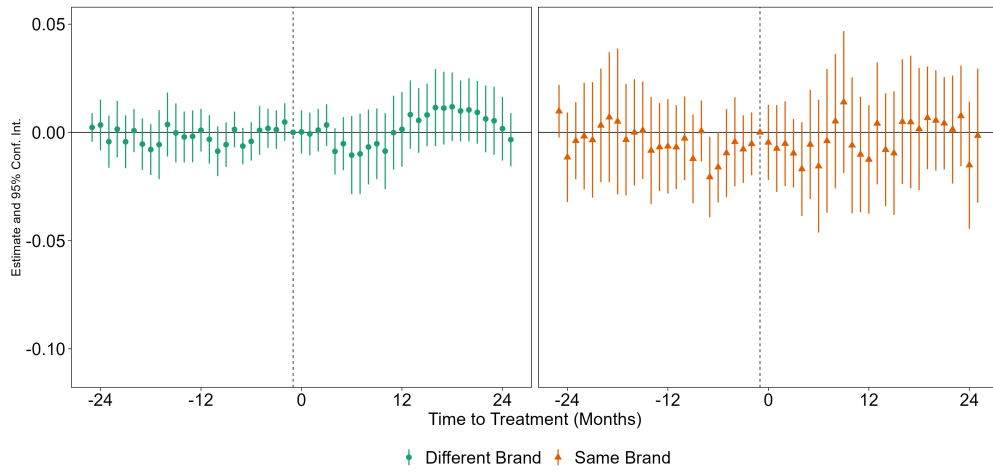
(b) Station Exit

Notes: Graph plots the coefficient estimates and 95% confidence intervals for the effect of station entry (in panel (a)) and exit (in panel (b)) within 1 mile of an incumbent station on incumbent pricing. Results are plotted separately for incumbent stations that share and do not share the same store brand as the entering or exiting station at the time of the event. Regressions are estimated using the two-stage DID estimator in [Gardner et al. \(2024\)](#). Standard errors are clustered at the city level. Data are sourced from the Oil Price Information Service (OPIS) for all stations in California for 2014-2018.

Appendix Figure A9: Effect of Station Entry and Exit, by Gasoline Branding



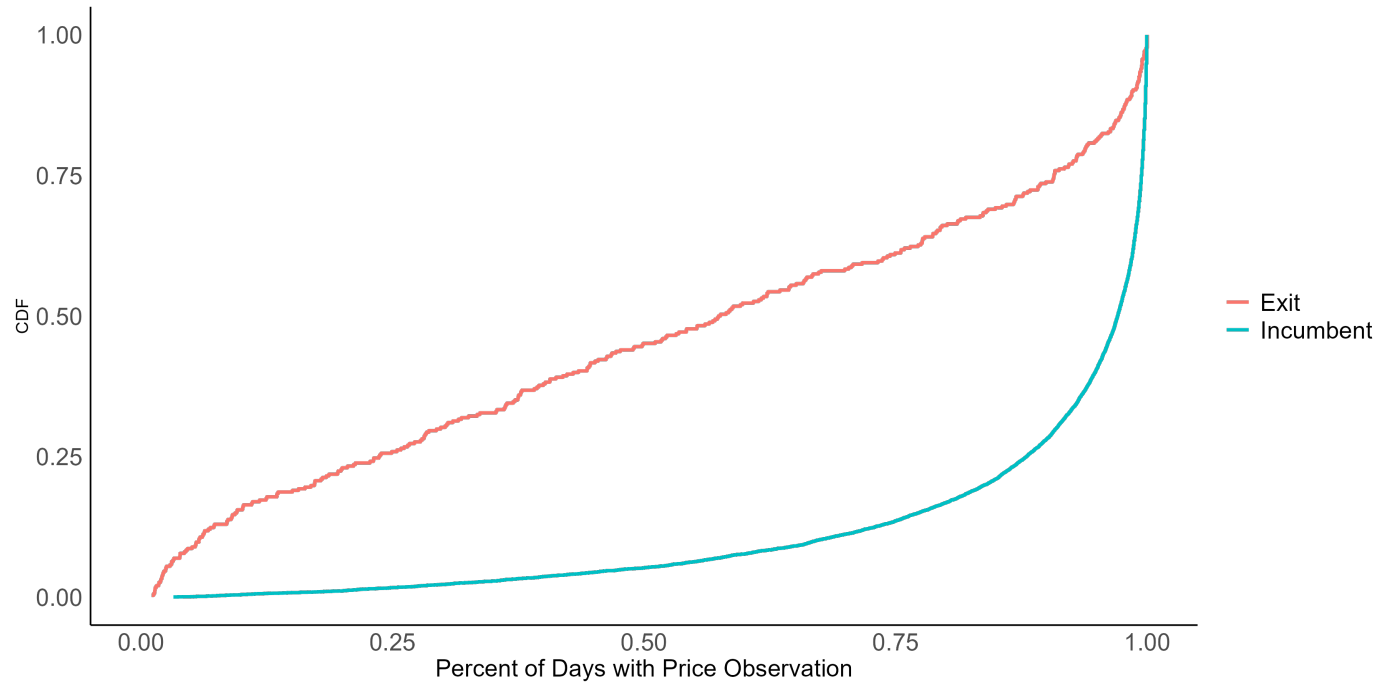
(a) Station Entry



(b) Station Exit

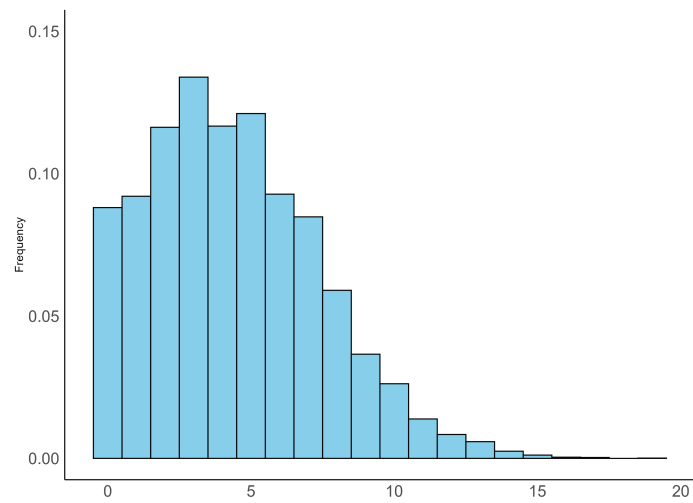
Notes: Graph plots the coefficient estimates and 95% confidence intervals for the effect of station entry (in panel (a)) and exit (in panel (b)) within 1 mile of an incumbent station on incumbent pricing. Results are plotted separately for incumbent stations that share and do not share the same gasoline brand as the entering or exiting station at the time of the event. Regressions are estimated using the two-stage DID estimator in [Gardner et al. \(2024\)](#). Standard errors are clustered at the city level. Data are sourced from the Oil Price Information Service (OPIS) for all stations in California for 2014-2018.

Appendix Figure A10: CDF of Frequency of Reported Prices



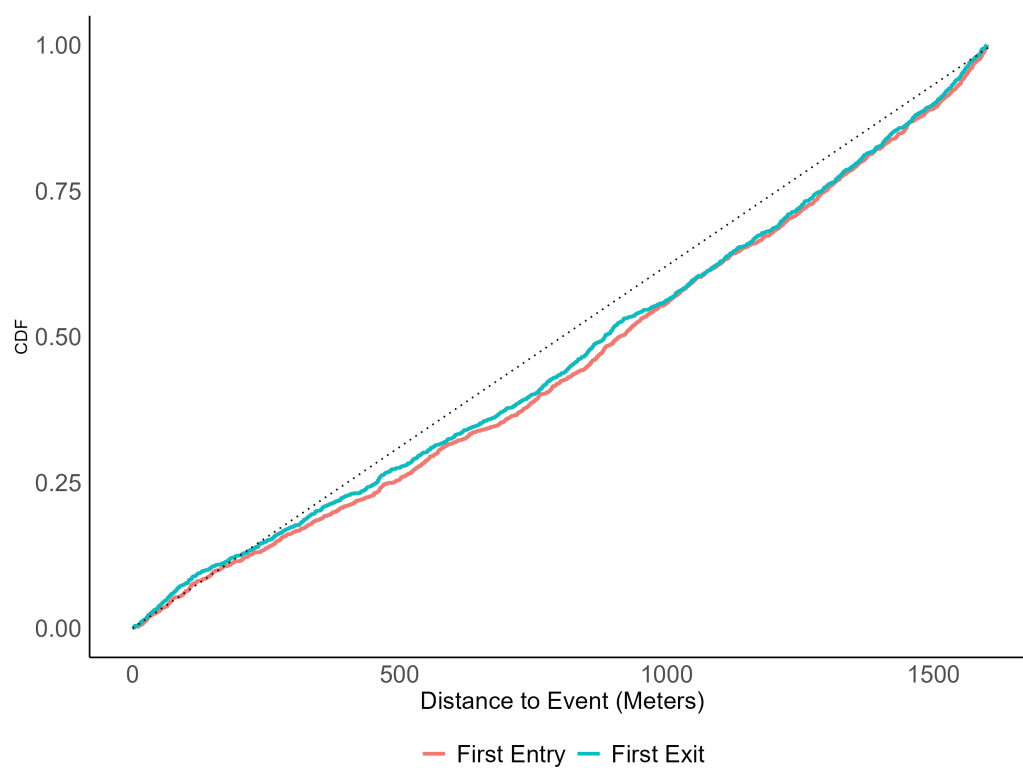
Notes: Graph plots The cumulative distribution function for the percent of potential days a station has a valid price reported in the OPIS data. The percentage is calculated as the number of price observations divided by the number of days spanned between the first and last price observation by station. Exit stations exit at some point during our sample. Incumbents neither exit nor enter during the sample period. Data are sourced from the Oil Price Information Service (OPIS) for all stations in California for 2014-2018.

Appendix Figure A11: Initial Market Size, by Station



Notes: Graph plots the number of other stations within 1 mile on the date of the earliest reported price observation by station. Data are sourced from the Oil Price Information Service (OPIS) for all stations in California for 2014-2018.

Appendix Figure A12: CDF of Distance from Incumbent Station to Entrant/Exiting Station



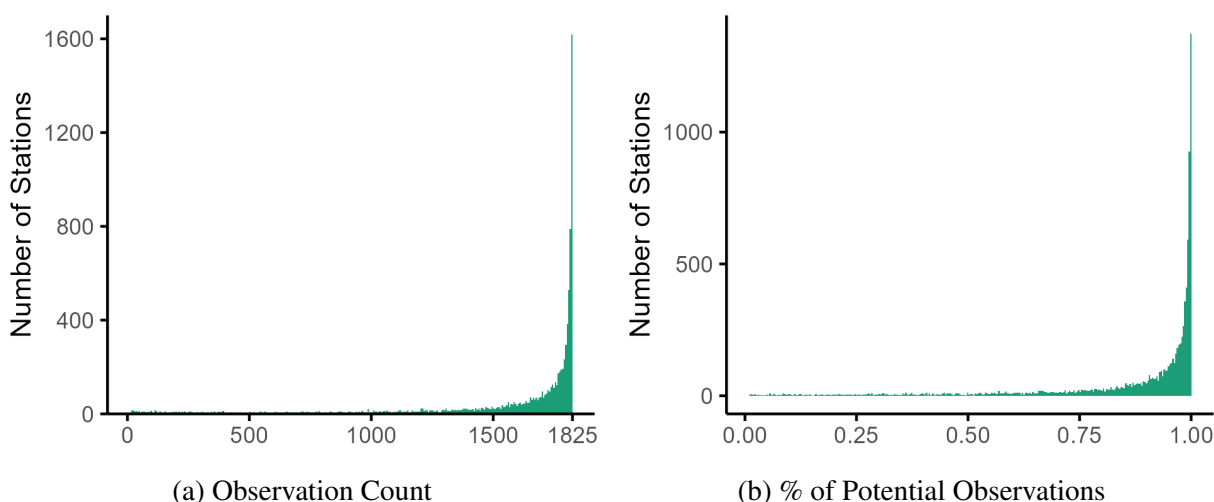
Notes: Graph plots the cumulative distribution function for the distance from the incumbent station to the entrant or exiting station. Data are sourced from the Oil Price Information Service (OPIS) for all stations in California for 2014-2018.

B Appendix: Data

This appendix provides additional detail on the construction and coverage of the OPIS gasoline price data and the procedures used to identify station entry and exit. Because our empirical design relies on high-frequency station observations, we document reporting completeness, potential sources of misclassification, and the spatial and temporal patterns of observed entry and exit.

The station-level panel of gasoline prices includes 9,539 stations with more than 35 million price observations across the three major gasoline grades: regular, mid-grade, and premium. However, prices are not observed every day for each station or fuel blend. Panel (a) of Figure A13 shows the distribution of observation counts by station over the 2014–2018 sample (1,825 days) for regular gasoline. To account for variation in station reporting windows, Panel (b) reports observations as a share of potential reporting days, defined as the number of days between a station’s first and last observed prices. The median station reports prices on 97% of days in its sample window, and 75% of stations report prices on more than 86% of potential days.

Appendix Figure A13: Price Observations, by Station



Notes: Panel (a) reports the number of observations by gas station over the sample period. Panel (b) scales the number of observations by the number of days between a station’s first and last observations. Data are sourced from the Oil Price Information Service (OPIS) for all stations in California for 2014–2018.

Classifying entry and exit using first and last observed prices introduces three potential sources of measurement error. First, the recorded entry or exit date may not perfectly align with the true start or end of operations. Given the high frequency and multiple data collection channels in OPIS (transaction data, crowd-sourced reports, and brand partnerships), we expect any resulting bias to be limited. Around the estimated entry or exit date, the median affected station reports prices on 78% of days.

Second, because prices are not reported daily for every station, some continuing stations may appear to enter after January 1, 2014 or exit before December 31, 2018. Such misclassification would attenuate estimated effects. To reduce false positives, we classify stations as entrants only if their first price occurs after March 1, 2014 and as exiting only if their last price occurs before November 1, 2018.³³ Under this definition, we identify 484 entry events and 348 exits.

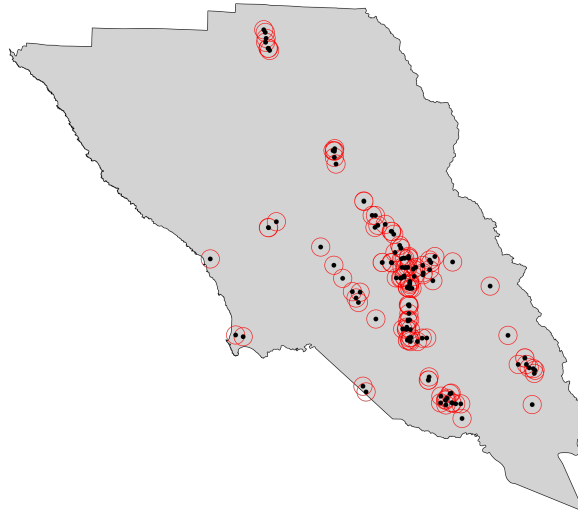
Third, some stations may never appear in OPIS. To assess coverage, we benchmark OPIS counts against the Census Bureau's County Business Patterns and California Energy Commission (CEC) estimates. The County Business Patterns data reports the number of businesses as of the week of March 12th of the appropriate year broken out by NAICS code. The data show a net increase of 228 gas stations in California from March 2014-March 2019. The CEC also undertakes an effort to estimate the number of stations in California based on returns from the A15 survey and other government data sources. For the same period, the CEC estimates a net increase of 190 stations in the state and around 10,000 gas stations in total, closely aligned with OPIS totals.

B.1 Patterns of Entry and Exit

Using station entry and exit dates and geographic coordinates, we define markets as one-mile radii around each station and compute the number of nearby competitors. Appendix Figure A14 illustrates the one-mile market definition in Sonoma County. We also report results using other distance measures, varying from .5 to 10 miles.

³³Using six-month buffers yields qualitatively similar results.

Appendix Figure A14: Map of Gasoline Stations in Sonoma County: 2014-2018

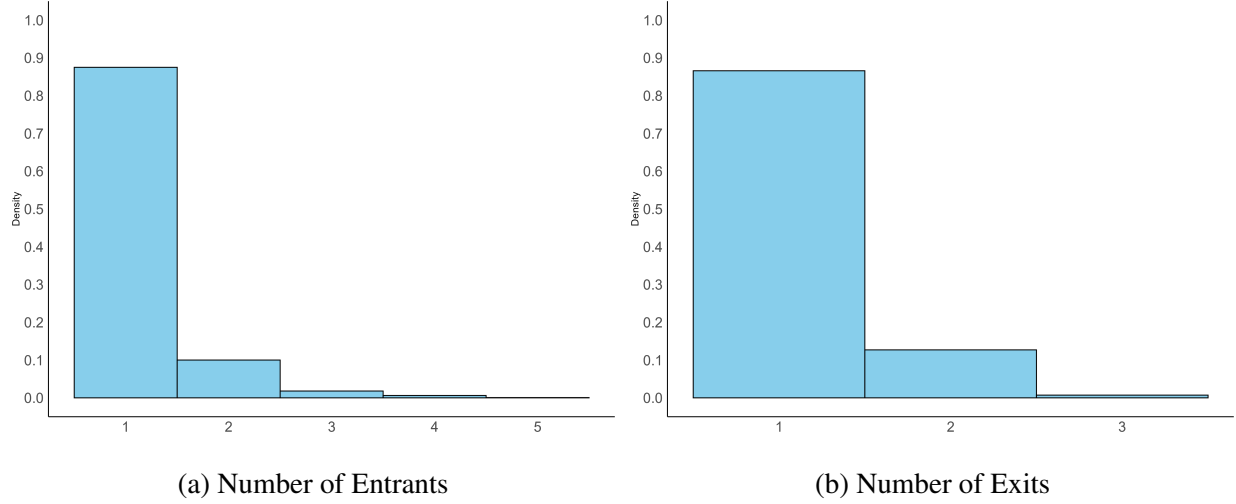


Notes: Station locations in Sonoma county, California are shown in black and their 1-mile market definition are shown in red. Data are sourced from the Oil Price Information Service (OPIS) for all stations in California for 2014-2018.

Panel (a) of Figure [A15](#) shows the number of entry events experienced by incumbents (conditional on at least one entry). While 85% of stations experience no nearby entry, multiple entries are rare: among affected stations, 88% experience only one entrant. Similarly, in Panel (b), while 85% of stations do not experience a nearby exit, 87% of those that do, only experience a single exit. To avoid contamination from multiple shocks, the main analysis focuses on the first observed entry or exit.

Appendix Figure [A11](#) reports the distribution of initial market sizes. Markets contain relatively few stations, with the median store competing against four other gas stations. Although some stations operate in monopoly markets and one station in downtown Los Angeles faces 19 competitors, only 10% of stations operate in markets with eight or more rivals. Figure [A12](#) plots the distribution of distances between incumbents and the first entering or exiting station within one mile; distances are approximately uniformly distributed.

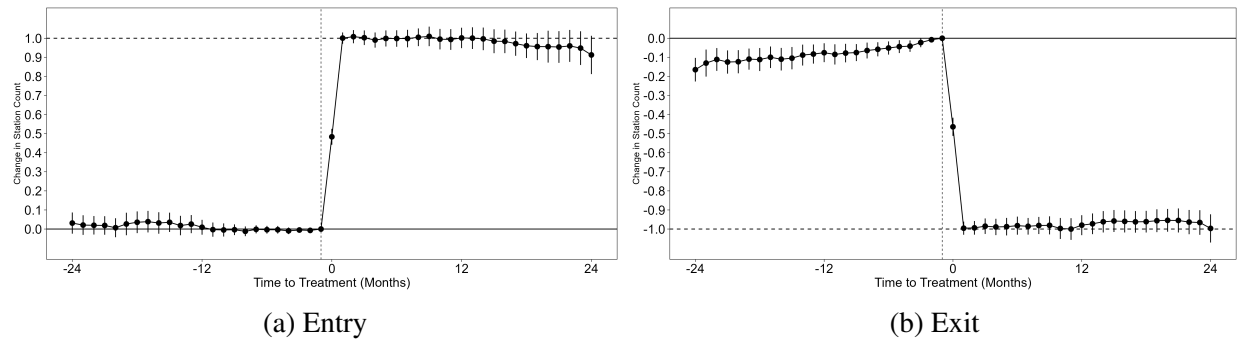
Appendix Figure A15: Number of Nearby Entries and Exits, by Station



Notes: Panel (a) reports the number of entry events per incumbent station, conditional on experiencing at least 1 entry event. Panel (b) reports the number of exit events per incumbent station, conditional on experiencing at least 1 exit event. Data are sourced from the Oil Price Information Service (OPIS) for all stations in California for 2014-2018.

Our difference-in-differences and event-study estimates would be attenuated if entry merely replaced a nearby exiting firm (or vice versa), leaving market structure unchanged. To assess this possibility, we estimate event-study models of the number of competitors within one mile around entry and exit events. Figure A16 reports the results. Prior to entry, competitor counts are stable, providing no evidence of differential pre-trends. At the time of entry, the number of nearby stations increases by approximately one and remains elevated over the following year, indicating that entry typically represents a net market expansion rather than replacement. Exit dynamics are similar. We find little evidence of offsetting entry around exit events. These patterns support the interpretation that observed entry and exit events reflect meaningful changes in local competition.

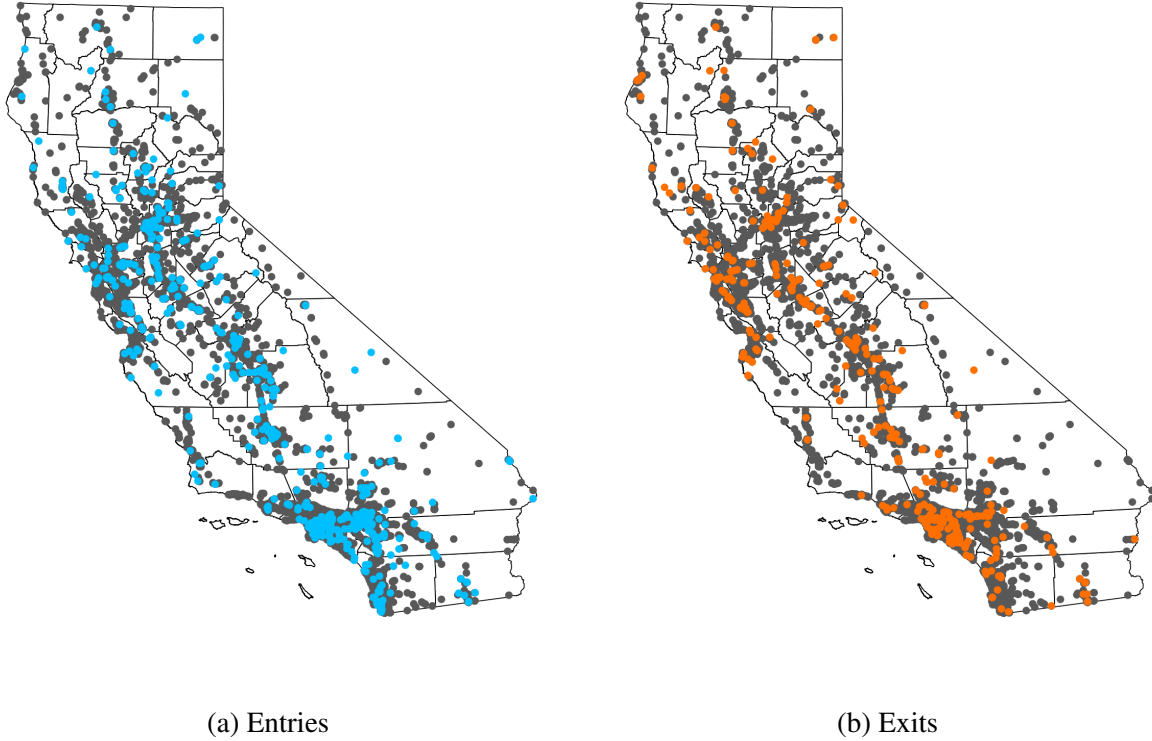
Appendix Figure A16: Effect of Station Entry and Exit on the Number of Competitors



Notes: Graph plots coefficient estimates and 95% confidence intervals for the effect of station entry (panel (a)) and exit (panel (b)) within 1 mile of an incumbent station on the number of competitors within 1 mile. Coefficients are estimated from a linear regression of station count within 1 mile on a panel of event time dummy variables, day-of-sample fixed effects, and station fixed effects. Standard errors are clustered at the city level. Data are sourced from the Oil Price Information Service (OPIS) for all stations in California for 2014-2018.

Geographically, entry and exit occur throughout California. Figure A17 maps entrant stations (blue) and exiting stations (orange). Although events are more common in urban areas and along the I-5 and CA-99 corridors, entry broadly mirrors the existing station footprint rather than concentrating solely in new markets. Entry is also observed in rural areas, though these events contribute to identification only when located sufficiently close to incumbent stations.

Appendix Figure A17: Map of Station Entry and Exit



Notes: Entrant stations are shown in panel (a) by blue dots. Exiting stations are represented by orange dots in panel (b). Grey dots reflect incumbent stations that neither enter nor exit during the sample. Data sourced from the Oil Price Information Service (OPIS) for all stations in California from 2014-2018.

B.2 Misclassification of Exiting Stations

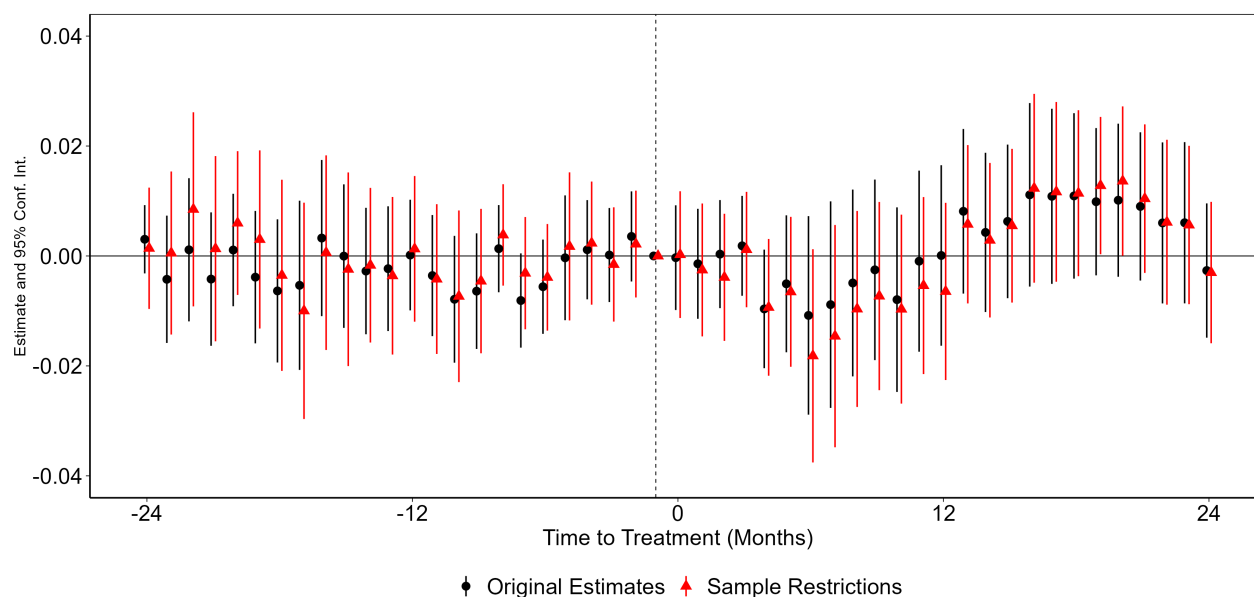
As a robustness check to our main results shown in Figure 2, we evaluate whether misclassification of exiting stations might provide an explanation for the null results we observe for exit events. As discussed above, we impute the entry and exit dates of stations from the dates of the first and last price observations in the data for each station. If a station has highly sporadic price observations, we could misclassify the first or last observation as an entry or exit, when in reality they had subsequent observations outside the bounds of our sample.

To test this, we make two sample restrictions to increase our confidence that misclassification is not driving the observed null results for exit events. First, we no longer consider any station that has less than 25% of potential days with a reported price as an exit. Secondly, we originally did

not consider first price observations in the first 3 months or last observations in the last 2 months of the sample to be entries or exits, and instead considered these stations to be active throughout the entire sample. We increase that restriction to no longer consider entries and exits in the first and last 6 months of the sample. As an example, a station whose last price is reported in August 2018 will no longer be considered an exiting station and will not create an exit event for nearby incumbents.

In Figure A18 we find little difference in our estimates after excluding the exiting stations for which misclassification might be the most prevalent. We continue to find a null effect of exit on incumbent station pricing.

Appendix Figure A18: Effect of Station Exit on Incumbent Pricing



Graph plots coefficients and 95% confidence intervals for the effect of station exit within 1 mile of an incumbent station on incumbent pricing. Estimates from the main results in Figure 2b are shown in black. The specification with additional sample restrictions is shown in red and excludes exiting stations that report a price on less than 25% of days and exit events in the final 6-months of the sample. Regressions are estimated using the two-stage DID estimator in [Gardner et al. \(2024\)](#). Standard errors are clustered at the city level. Data are sourced from the Oil Price Information Service (OPIS) for all stations in California for 2014-2018.

C Appendix: Model

Consider a model in which K single-product firms set prices for differentiated products. Firm k faces constant marginal costs of c_k and the utility of product i to consumer j is given by

$$u_{ij} = \delta_i - \alpha p_i + \varepsilon_{ij} \quad (6)$$

where δ_i captures the mean utility offered by good i , α is sensitivity of utility to the price of the good, p_i , and ε_{ij} reflects the idiosyncratic utility enjoyed by consumer j when choosing good i . Normalizing the utility of the outside good to 0 and under the standard logit assumption that ε_{ij} is distributed IID, Extreme Value Type I, the market share, s_i , of good i is given by

$$s_i = \frac{e^{\delta_i - \alpha p_i}}{1 + \sum_k e^{\delta_k - \alpha p_k}}, \quad (7)$$

and firm i earns profits given of

$$\Pi_i = (p_i - c_i)Ms_i - F, \quad (8)$$

where M is the exogenously determined size of market (i.e., number of potential consumers) and F are fixed costs.

Maximizing profits with respect to p_i , we have

$$\begin{aligned} \frac{\partial \Pi_i}{\partial p_i} &= Ms_i + (p_i - c_i)M \frac{\partial s_i}{\partial p_i} \\ &= Ms_i - (p_i - c_i)M(\alpha s_i(1 - s_i)) \end{aligned} \quad (9)$$

which yields the first-order condition for profit maximization for firm i :

$$p_i = c_i + \frac{1}{\alpha(1 - s_i)} \quad (10)$$